Biomedical Sciences

Vrije Universiteit Amsterdam - Faculty of Science - Biomedical Sciences - 2025-2026

The key objective of the M Biomedical Sciences is to train students to conduct research on a wide range of biomedical issues. Biomedical science is increasingly becoming an interdisciplinary research field with different disciplines providing diverse knowledge about the biomedical issue and interaction with society, which is required to effectively address the biomedical issue. Therefore, the program equips students with knowledge, skills and competences to work from and integrate knowledge from different biomedical disciplines.

The program builds on the knowledge, research methodologies, and skills from an undergraduate scientific background by first deepening knowledge, skills and understanding in biomedical disciplinary research (either Immunology, Infectious Diseases, or Neurobiology) and second by broadening this knowledge, either by acquiring knowledge, skills and understanding of a second biomedical discipline or by broadening the skills and competences to conduct biomedical research at the interface with society (either International Public Health (IPH), Science Communication, Science in Society or Education) -students who do not pursue a second specialization of the program must take a coherent set of biomedical courses from another discipline. Through this arrangement of the program, each student specializes in 2 disciplines.

The programme has three learning goals :

- Understanding the integration of various biomedical disciplines and the societal interface in order to address biomedical issues.
- Deeping knowledge, skills and competences in a specific biomedical discipline
- Broadening knowledge, skills and competences to another biomedical discipline or do research on the interface of science and society

It is important to show students that biomedical specializations are not isolated disciplines, but instead are integrated with each other and transcend disciplinary boundaries. Therefore, the program starts with a compulsory “Integrated Biomedical Sciences” course . All students participate in this course, which aims to present the interdisciplinary nature of biomedical sciences.

In parallel with and after this course, students take 3 courses from a biomedical disciplinary specialization. The biomedical disciplinary specializations include mostly lectures and working groups to gain knowledge of current theory and key research questions in the field. In the second semester, the students are engaged in an individual research internship within the respective discipline.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Biomedical Sciences year 1

Description

In the first year of the M Biomedical Sciences, students choose from one of the three biomedical disciplinary research specializations. Next to the specialization, students also follow some general compulsory courses.

Biomedical disciplinary research specializations:

- Immunology
- Infectious Diseases
- Neurobiology (maximum number of students allowed)

Programme Sections

- Biomedical Sciences year 1 compulsory courses
- Biomedical Sciences research specialisations year 1 and/or 2

Biomedical Sciences year 1 compulsory courses

Description

During the first year, students take four compulsory courses:

- Integrated Biomedical Sciences (6 EC), the general start course.
- Study and Career (0 EC), an academic year course (followed by a second (3 EC) course in the second year).
- Ethics (3 EC), part time compulsory course in period 3.
- Academic Writing (3 EC), part time course during the internship period.

Courses

Course Name	Period	Credits	Code
Study and Career year 1 M BMED	Ac. Year (September)	0EC	AM_1258
Integrated Biomedical Sciences	P1	6EC	AM_1281
Ethics in Life Sciences	P3	3EC	AM_470707
Academic Writing: Biomedical Sciences	P4+5+6	3EC	AM_1320

Biomedical Sciences research specialisations year 1 and/or 2

Description

First and/or second year biomedical disciplinary research specialisations:

- Immunology
- Infectious Diseases
- Neurobiology (maximum number of students allowed)

The prescribed scope of a research specialisation is 48 EC.

A research specialisation therefore includes:

- 3 courses from the specialisation (18 EC), including all compulsory specialisation courses
- a research internship (30 EC)*

* In some cases, students are allowed an extension of the internship (3-6 EC), the total EC for both internships together may never exceed 66 EC.

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

Programme Sections

- Biomedical Sciences specialisation Immunology
- Biomedical Sciences specialisation Infectious Diseases

- Biomedical Sciences specialisation Neurobiology

Biomedical Sciences specialisation Immunology

Description

The prescribed scope of the specialisation Immunology is 48 EC.

This research specialisation therefore includes:

- 3 compulsory courses from the specialisation (18 EC)
- a research internship (30 EC)

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

The Master's graduate with a specialisation in Immunology has a broad understanding of immunological processes, ranging from the molecular and cellular interactions, to an integrative knowledge of the role of the immune system in various pathologies, such as cancer, infectious diseases and autoimmunity. The Master's graduate has specialised in one of the subjects within the field of immunology. The Master's graduate possesses knowledge of current theory and the key research questions in the field of immunology and has an understanding of the scientific and social relevance of this subject area.

Programme Sections

- Biomedical Sciences specialisation Immunology compulsory courses

Biomedical Sciences specialisation Immunology compulsory courses

Courses

Course Name	Period	Credits	Code
Internship Immunology	Ac. Year (September)	30EC	AM_471137
Advanced Molecular Immunology and Cell Biology	P1	6EC	AM_470656
Clinical Immunology and Disease Mechanisms	P2	6EC	AM_1309
Immune Modulation, Monitoring and Metabolism	P2	6EC	AM_1318

Biomedical Sciences specialisation Infectious Diseases

Description

The prescribed scope of the specialisation Infectious Diseases is 48 EC.

This research specialisation therefore includes:

- 3 courses from the specialisation (18 EC), including all compulsory specialisation courses
- a research internship (30 EC)

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

The Master's graduate with a specialisation in Infectious diseases has a broad understanding of the biology of pathogenic organisms and the interaction between pathogens and their hosts. The Master's graduate has the ability to conduct scientific research in the field of medical microbiology and to critically assess the results of microbial research. The Master's graduate has specialised in one of the subjects within the field of medical microbiology. The Master's graduate possesses knowledge of current theory and the key research questions in this field and has an understanding of the scientific and social relevance of this subject area.

Programme Sections

- Biomedical Sciences specialisation Infectious Diseases constrained choice
- Biomedical Sciences specialisation Infectious Diseases compulsory courses

Biomedical Sciences specialisation Infectious Diseases constrained choice

Description

Students should choose 6 EC of these courses.

Courses

Course Name	Period	Credits	Code
Containment Strategies of Infectious Diseases in Global Context	P1	6EC	AM_470127
Parasitology	P2	6EC	AM_470052
The Human Microbiome in Health and Disease	P3	6EC	AM_1310

Biomedical Sciences specialisation Infectious Diseases compulsory courses

Courses

Course Name	Period	Credits	Code
Internship Infectious Diseases	Ac. Year (September)	30EC	AM_471138
Advanced Molecular Immunology and Cell Biology	P1	6EC	AM_470656
Molecular Infection Biology	P2	6EC	AM_470657

Biomedical Sciences specialisation Neurobiology

Description

The prescribed scope of the specialisation Neurobiology is 48 EC.

This research specialisation therefore includes:

- 3 compulsory courses from the specialisation (18 EC)
- a research internship (30 EC)

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

The Master's graduate with a specialisation in Neurobiology has knowledge, insight and understanding of the state of the art in terms of developing theories and insight into the most important current research issues in the neurosciences. The Master's graduate has the ability to conduct scientific research in the field of neurobiological research and to critically assess the results. The Master's graduate has specialized in one of the subjects within the field of neurobiology. The Master's graduate possesses knowledge of the significance of neurobiology within the context of brain research and some of its clinical implications.

Programme Sections

- Biomedical Sciences specialisation Neurobiology compulsory courses

Biomedical Sciences specialisation Neurobiology compulsory courses

Courses

Course Name	Period	Credits	Code
Internship Neurobiology	Ac. Year (September)	30EC	AM_1178
From Molecule to Mind	P1	6EC	AM_1275
Clinical neurosciences	P2	6EC	AM_1005
Genetics in Neuroscience	P2	6EC	AM_1214

Biomedical Sciences year 2

Description

After the first year biomedical disciplinary research specialization, the programme is completed with a second

specialization and the compulsory second year courses.

The second specialization allows the students to choose their profile:

- Research: choose one of the three biomedical disciplinary research specializations, different from the first year
- Societal: choose an ICSE specialization
- Research Focus Program: free curriculum with second research internship (coherent package of courses, different from first year specialization)

Programme Sections

- Biomedical Sciences year 2 constrained choice
- Biomedical Sciences year 2 electives
- Biomedical Sciences year 2 specialisations
- Biomedical Sciences year 2 compulsory courses

Biomedical Sciences year 2 constrained choice

Description

The student must choose one of the following three options for the mandatory literature study:

Literature Thesis M Bmed (AM_471135, 9EC): Individual project supervised by a VU supervisor in the field of Immunology, Infectious Diseases, or Neurobiology (depending on your chosen specialisation(s)). Self-arranged, flexible scheduling (6 weeks full-time/12 weeks part-time).

Literature Study M Bmed (AM_1391, 6EC): Group project supervised by assigned supervisors/topics. Support through workshops and content discussions with your supervisor. Period 1 ór Period 3.

Literature Review MPA (AM_1251, 6EC): Only for students in the specialisation International Public Health, Science Communication or Science in Society. Group project supervised by MPA-assigned supervisors/topics. Support through workshops and content discussions with your supervisor, including an exam. Period 2.

Courses

Course Name	Period	Credits	Code
Literature thesis Biomedical Sciences	Ac. Year (September)	9EC	AM_471135
Literature Review MPA	P1, P2	6EC	AM_1251
Literature Study M Biomedical Sciences	P1, P3	6EC	AM_1319

Biomedical Sciences year 2 electives

Courses

Course Name	Period	Credits	Code
Advanced Molecular Immunology and Cell Biology	P1	6EC	AM_470656
Analysis of Governmental Policy	P1	6EC	AM_470571
Containment Strategies of Infectious Diseases in Global Context	P1	6EC	AM_470127
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587
Business Management in Health and Life Sciences	P2	6EC	AM_470584
Clinical Immunology and Disease Mechanisms	P2	6EC	AM_1309
Communication, Organization and Management	P2	6EC	AM_470572
Health, Globalisation and Human Rights	P2	6EC	AM_470818
Immune Modulation, Monitoring and Metabolism	P2	6EC	AM_1318
International Comparative Analyses of Health Care Systems	P2	6EC	AM_470820
Management of Innovative Technologies in Community-Based Health Care	P2	6EC	AM_1181
Molecular Infection Biology	P2	6EC	AM_470657
Parasitology	P2	6EC	AM_470052
Policy, Management and Organisation in International Public Health	P2	6EC	AM_470819

Course Name	Period	Credits	Code
Policy, Politics and Participation	P2	6EC	AM_470589
Science Journalism	P2	6EC	AM_471014
Science Museology	P2	6EC	AM_470590
Clinical Development and Clinical Trials	P3	3EC	AM_1180
Disability and Development	P3	6EC	AM_470588
Epidemiology	P3	3EC	AM_1179
Innovation, Behavior, Emergence and Markets	P3	3EC	AM_1052
Science in Dialogue	P3	6EC	AM_1002
The Human Microbiome in Health and Disease	P3	6EC	AM_1310

Biomedical Sciences year 2 specialisations

Description

Choose your profile:

- Research: choose one of the three biomedical disciplinary research specializations, different from the first year
- Societal: choose an ICSE specialization
- Research Focus Program: free curriculum with second research internship (coherent package of courses, different from first year specialization)

Programme Sections

- Biomedical Sciences Research focus program
- Biomedical Sciences ICSE specialisations year 2
- Biomedical Sciences research specialisations year 1 and/or 2

Biomedical Sciences Research focus program

Description

The prescribed scope of the research focus programme is 48 EC.

The research focus programme therefore includes:

1. A minimum of 18 EC consisting of:

- a group of coherent courses at Master's level with biomedical focus, not related to the first year research specialisation (12-18 EC)
- an extension of the internship (3-6 EC)*

2. a research internship (30 EC) in a biomedical topic*

* *The total EC for both internships together may not exceed 66 EC*

Please note the contents of the Research Focus Program need approval (of the Examination Board) via the master's coordinator.

The programme is completed with the compulsory general courses (24 EC) and a first year research specialisation of at least 48 EC.

Courses

Course Name	Period	Credits	Code
Internship Biomedical Sciences	Ac. Year (September)	30EC	AM_471158

Biomedical Sciences ICSE specialisations year 2

Description

The I/C/S/E specialisations are only available in the second year of the master:
International Public Health (IPH; 48 EC)
Science Communication specialisation (Comm: 48 EC)
Specialisation Science in Society (SiS; 48 EC)
Education specialisation (Edu: 60 EC)

The prescribed scope of the specialisations in IPH, Communication or SiS is 48 EC, and includes:

- at least 3 courses from the specialisation (18 EC)
- an internship within the specialisation (30 EC)

The Master's programme is completed with a first year research specialisation (48 EC) and the compulsory general courses (24 EC).

Course registration

Students may participate in the second year specializations Science in Society or Science Communication only if the student passed 24EC of their first year.

Programme Sections

- Biomedical Sciences ICSE specialisation Education
- Biomedical Sciences ICSE specialisation International Public Health
- Biomedical Sciences ICSE specialisation Science Communication
- Biomedical Sciences ICSE specialisation Science in Society

Biomedical Sciences ICSE specialisation Education

Description

The Master’s graduate with a specialization in Education obtains a certificate that qualifies the graduate to teach Biology in secondary schools (this is a ‘grade one’ certificate, i.e. it qualifies the graduate to teach pupils who will sit public exams in the subject).

The programme for the Education specialization essentially consists of one year of specific teacher training. This 60 EC-programme is taught in Dutch. Note that the Education Specialization is identical to the Master’s in ‘Leraar Voorbereidend Hoger Onderwijs - Biologie’. The programme can be started twice a year, in September and February.

For courses and more information on the Education specialization: https://vu.nl/nl/onderwijs/master/leraar-vho-in-de-betawetenschappen/

The prescribed scope of the Education specialization is 60 EC

Course registration

Apply before 1 May (start Sep) or 1 Nov (start Feb).

Programme Sections

- Secondary Education Teacher in Science - Specialization Biology

Secondary Education Teacher in Science - Specialization Biology

Description

Heb jij een open blik en ben je nieuwsgierig naar wat jongeren beweegt en bezighoudt? Vind je het leuk om jouw kennis over Biologie over te dragen op leerlingen en wil jij getuige zijn van hun persoonlijke ontwikkeling? Biologie gaat over het begrijpen van jezelf en de wereld om je heen. Als leraar Biologie leer je de leerlingen meer over wie ze zijn en wat hun plek is in de wereld. Zo kunnen ze zorgdragen voor hun eigen gezondheid én voor hun omgeving. Dat doe je door in jouw lessen thema’s als transport, neurale regulatie en evolutie te behandelen, maar ook aandacht te geven aan sociaalmaatschappelijke vraagstukken. Kortom, docent Biologie is een heel veelzijdige baan!

Courses

Course Name	Period	Credits	Code
Schoolvak Bio Ecologie en biodiversiteit	Ac. Year (September)	5EC	O_ABBECBD
Schoolvak Bio Evolutie	Ac. Year (September)	3EC	O_ABBIEVO
Schoolvak Bio Planten	Ac. Year (September)	5EC	O_ABBIPL

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2

Course Name	Period	Credits	Code
Practice 2	P2+3	9EC	O_MLPRAK_2
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Biomedical Sciences ICSE specialisation International Public Health

Description

The specialisation International Public Health is only available in the second year of the master.

The prescribed scope of the specialisation is 48 EC, and includes:

- the 3 compulsory courses from the specialisation (18 EC)
- an internship within the specialisation (30 EC)

The Master's programme is completed with a first year research specialisation (48 EC) and the compulsory general courses (24 EC, including the literature thesis within the field of the first year research specialisation).

A background in epidemiology is compulsory to be admissible to this specific specialisation (for entry in the course Research Methods).

The Master’s graduate with a specialisation in International public health has a broad understanding of current and future challenges in international public health, their main causes, and applied and potential interventions. The Master’s graduate has specialised knowledge of relevant concepts from various disciplines, including epidemiology, policy science, anthropology, management studies, biomedical sciences and health sciences. The Master’s graduate has the ability to conduct scientific research in the field of international public health and to critically assess the results of international public health research. The Master’s graduate has specialised in one of the subjects within the field of international public health. The Master’s graduate possesses knowledge of current theory and the key research questions in this field and has an understanding of the scientific and social relevance of this subject area.

Programme Sections

- Biomedical Sciences specialisation International Public Health compulsory courses

Biomedical Sciences specialisation International Public Health compulsory courses

Course registration

You need to have a background in Epidemiology to be able to do this specialization. See the course 'Research

methods' for more information.

Courses

Course Name	Period	Credits	Code
Internship International Public Health	Ac. Year (September)	30EC	AM_471139
Containment Strategies of Infectious Diseases in Global Context	P1	6EC	AM_470127
Research Methods for Health Sciences	P1	6EC	AM_1255
Policy, Management and Organisation in International Public Health	P2	6EC	AM_470819

Biomedical Sciences ICSE specialisation Science Communication

Description

The specialisation Science Communication is only available in the second year of the master.

The prescribed scope of the specialisation is 48 EC, and includes:

- 3 courses from the specialisation (18 EC), including all compulsory courses
- an internship within the specialisation (30 EC)

The Master's programme is completed with a first year research specialisation (48 EC) and the compulsory general courses (24 EC, including the literature thesis within the field of the first year research specialisation).

Biomedical science is increasingly becoming an interdisciplinary research field in which biomedical scientists can no longer function effectively in isolation. Rather, they benefit from interaction with other scientists (such as those in the fields of molecular biology, neurobiology and immunobiology) and societal actors (such as doctors, patients and policymakers). Communication about science takes place between academic peers and between scientists and the general public. This makes the Communication specialisation a complex and dynamic field of research and practice, for example on patient participation in health research, the use and effects of media metaphors and hype, and public understanding of emergent technologies. The Master’s graduate with this specialisation has a theoretical understanding of the complex problems that arise during such communication processes, and has developed the necessary skills to act professionally at this interface to enhance communication and the outcomes of communication between scientific actors and society.

Programme Sections

- Biomedical Sciences ICSE specialisation Science Communication compulsory courses
- Biomedical Sciences ICSE specialisation Science Communication constrained choice
- Biomedical Sciences ICSE specialisation Science Communication constrained choice - internship

Biomedical Sciences ICSE specialisation Science Communication compulsory courses

Courses

Course Name	Period	Credits	Code
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587

Biomedical Sciences ICSE specialisation Science Communication constrained choice

Description

The student should choose one (6 EC) of these courses.

Courses

Course Name	Period	Credits	Code
Science Journalism	P2	6EC	AM_471014
Science Museology	P2	6EC	AM_470590

Course Name	Period	Credits	Code
Science in Dialogue	P3	6EC	AM_1002

Biomedical Sciences ICSE specialisation Science Communication constrained choice - internship

Description

The student should choose one of these courses.

Courses

Course Name	Period	Credits	Code
Reflective Practice Internship Science Communication	Ac. Year (September)	30EC	AM_1163
Research Internship Science Communication	Ac. Year (September)	30EC	AM_1162

Biomedical Sciences ICSE specialisation Science in Society

Description

The specialisation Science in Society is only available in the second year of the master.

The prescribed scope of the specialisation is 48 EC, and includes:

- the 3 compulsory courses from the specialisation (18 EC)
- an internship within the specialisation (30 EC)

The Master's programme is completed with a first year research specialisation (48 EC) and the compulsory general courses (24 EC).

The Master’s graduate with a specialisation Science in Society combines an academic approach with the skills and competences that will allow him or her to perform scientific research at the interface of the biomedical sciences and society. The specialisation aims to develop strategies that contribute to an understanding of complex societal problems and strategies to solve complex societal problems through interdisciplinary research. In addition, the programme analyses the social, economic and ethical aspects of new developments in the biomedical sciences, so as to assess their implications for society. Master’s graduates have the necessary skills to collaborate and communicate with researchers from various scientific disciplines (including but not limited to those in the life sciences) and societal actors, and the ability to use these academic insights.

Programme Sections

- Biomedical Sciences ICSE specialisation Science in Society compulsory courses

Biomedical Sciences ICSE specialisation Science in Society compulsory courses

Courses

Course Name	Period	Credits	Code
Internship Science in Society	Ac. Year (September)	30EC	AM_1133
Analysis of Governmental Policy	P1	6EC	AM_470571
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Communication, Organization and Management	P2	6EC	AM_470572

Biomedical Sciences research specialisations year 1 and/or 2

Description

First and/or second year biomedical disciplinary research specialisations:

- Immunology
- Infectious Diseases
- Neurobiology (maximum number of students allowed)

The prescribed scope of a research specialisation is 48 EC.

A research specialisation therefore includes:

- 3 courses from the specialisation (18 EC), including all compulsory specialisation courses
- a research internship (30 EC)*

* In some cases, students are allowed an extension of the internship (3-6 EC), the total EC for both internships together may never exceed 66 EC.

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

Programme Sections

- Biomedical Sciences specialisation Immunology
- Biomedical Sciences specialisation Infectious Diseases
- Biomedical Sciences specialisation Neurobiology

Biomedical Sciences specialisation Immunology

Description

The prescribed scope of the specialisation Immunology is 48 EC.

This research specialisation therefore includes:

- 3 compulsory courses from the specialisation (18 EC)
- a research internship (30 EC)

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

The Master's graduate with a specialisation in Immunology has a broad understanding of immunological processes, ranging from the molecular and cellular interactions, to an integrative knowledge of the role of the immune system in various pathologies, such as cancer, infectious diseases and autoimmunity. The Master's graduate has specialised in one of the subjects within the field of immunology. The Master's graduate possesses knowledge of current theory and the key research questions in the field of immunology and has an understanding of the scientific and social relevance of this subject area.

Programme Sections

- Biomedical Sciences specialisation Immunology compulsory courses

Biomedical Sciences specialisation Immunology compulsory courses

Courses

Course Name	Period	Credits	Code
Internship Immunology	Ac. Year (September)	30EC	AM_471137
Advanced Molecular Immunology and Cell Biology	P1	6EC	AM_470656
Clinical Immunology and Disease Mechanisms	P2	6EC	AM_1309
Immune Modulation, Monitoring and Metabolism	P2	6EC	AM_1318

Biomedical Sciences specialisation Infectious Diseases

Description

The prescribed scope of the specialisation Infectious Diseases is 48 EC.

This research specialisation therefore includes:

- 3 courses from the specialisation (18 EC), including all compulsory specialisation courses
- a research internship (30 EC)

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

The Master's graduate with a specialisation in Infectious diseases has a broad understanding of the biology of pathogenic organisms and the interaction between pathogens and their hosts. The Master's graduate has the ability to conduct scientific research in the field of medical microbiology and to critically assess the results of microbial research. The Master's graduate has specialised in one of the subjects within the field of medical microbiology. The Master's graduate possesses knowledge of current theory and the key research questions in this field and has an understanding of the scientific and social relevance of this subject area.

Programme Sections

- Biomedical Sciences specialisation Infectious Diseases constrained choice
- Biomedical Sciences specialisation Infectious Diseases compulsory courses

Biomedical Sciences specialisation Infectious Diseases constrained choice

Description

Students should choose 6 EC of these courses.

Courses

Course Name	Period	Credits	Code
Containment Strategies of Infectious Diseases in Global Context	P1	6EC	AM_470127
Parasitology	P2	6EC	AM_470052
The Human Microbiome in Health and Disease	P3	6EC	AM_1310

Biomedical Sciences specialisation Infectious Diseases compulsory courses

Courses

Course Name	Period	Credits	Code
Internship Infectious Diseases	Ac. Year (September)	30EC	AM_471138
Advanced Molecular Immunology and Cell Biology	P1	6EC	AM_470656
Molecular Infection Biology	P2	6EC	AM_470657

Biomedical Sciences specialisation Neurobiology

Description

The prescribed scope of the specialisation Neurobiology is 48 EC.

This research specialisation therefore includes:

- 3 compulsory courses from the specialisation (18 EC)
- a research internship (30 EC)

The programme is completed with the compulsory general courses (24 EC) and a second specialisation of 48 EC.

The Master's graduate with a specialisation in Neurobiology has knowledge, insight and understanding of the state of the art in terms of developing theories and insight into the most important current research issues in the neurosciences. The Master's graduate has the ability to conduct scientific research in the field of neurobiological research and to critically assess the results. The Master's graduate has specialized in one of the subjects within the field of neurobiology. The Master's graduate possesses knowledge of the significance of neurobiology within the context of brain research and some of its clinical implications.

Programme Sections

- Biomedical Sciences specialisation Neurobiology compulsory courses

Biomedical Sciences specialisation Neurobiology compulsory courses

Courses

Course Name	Period	Credits	Code
Internship Neurobiology	Ac. Year (September)	30EC	AM_1178
From Molecule to Mind	P1	6EC	AM_1275
Clinical neurosciences	P2	6EC	AM_1005
Genetics in Neuroscience	P2	6EC	AM_1214

Biomedical Sciences year 2 compulsory courses

Description

During the second year, students take one compulsory course:

Study and Career (3EC), which continues on from what students did in the first year Study and Career course.

Courses

Course Name	Period	Credits	Code
Study and Career year 2 M BMED	Ac. Year (September)	3EC	AM_1270

Courses

Academic Writing: Biomedical Sciences

Course Code	AM_1320
Credits	3
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G.A. Dreschler
Examiner	dr. G.A. Dreschler
Teaching Staff	dr. A. Martonfi, dr. G.A. Dreschler
Teaching method(s)	Study Group, Lecture

Course Objective

On successful completion of this course you will be a more confident writer in English in terms of producing an academic text which is coherent and linguistically correct. You will have insight into the most important stylistic aspects of academic texts in your discipline. You will also have insight into how specific linguistic structures contribute to text coherence, and will have the ability to identify and improve aspects of coherence in your own academic texts. Finally, you will have clear insight into your strengths and weaknesses of your academic English writing skills, and knowledge of how to further develop your strengths and reduce your weaknesses.

Course Content

The course will start off by ensuring that you are up to speed in terms of the macrostructural and discursive features of academic texts in general and the specific kind of text you are writing in the content of the Biomedical Sciences programme. Centre stage is then given to aspects of coherent text and to linguistic devices that English uses to express complex ideas: we introduce basic principles, give you practice in analysing and reformulating

defective text segments, and give you a checklist that you can use when editing your own text. The idea underlying this approach is that writing essentially involves rewriting and rewriting.

The treatment of linguistic accuracy (lexical, grammatical and punctuational features) takes a less central place in the group sessions of the course, but the detailed feedback given on your texts allows you to work on any individual problems you may have using the ELS-Online feedback site.

Additional Information Teaching Methods

Three lectures and three seminars in three weeks. Attendance is obligatory.

Method of Assessment

The assignments for this course are linked to the research report for your internship. You hand in two assignments for this course: (i) a version of your introduction and (ii) a version of your conclusion. You will receive feedback on the first assignment and will include a report on incorporating the feedback with the second assignment. Your second text will receive a pass/fail assessment. If you fail, you will be given the opportunity to submit a reworked version.

Literature

There is no obligatory reading for this course outside the course materials, but for language support you are advised to study the feedback categories in ELS-Online.

Advanced Molecular Immunology and Cell Biology

Course Code	AM_470656
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R.E. Mebius
Examiner	prof. dr. R.E. Mebius
Teaching Staff	dr. G. Kooij, dr. J.M.M. den Haan, prof. dr. R.E. Mebius, dr. S. Duinkerken, dr. F. van Maldegem, M.A.J.B. Schuitema, dr. ir. J. Van den Bossche, C.J. van Vliet MSc
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

To acquire insight into:

- the molecular and cellular aspects of lymphoid organ development
- cellular interactions within the immune system and how molecular diversity is generated to regulate immune responses.
- the various strategies of host immune responses against pathogens, and how pathogens escape proper immune responses.
- the various strategies of the host to positively or negatively affect immune responses during disease (cancer, inflammatory diseases, infections, such as COVID 19).
- mechanisms by which the immune system regulates either immune activation or tolerance induction.
- the mechanism of cell migration within the immune system.
- immunometabolism

End terms:

Knowledge: At the end of the course the student is familiar with current knowledge on the induction and regulation of immune responses in health and disease.

Skills:

- The student is capable of applying the acquired knowledge and can interpret scientific literature and scientific hypotheses of each of the topics described above.
- The student is able to read scientific literature, summarize the contents both at the level of the methods as well as the results section, and write a scientific abstract.

Course Content

Immunology is a rapid growing field of research in medicine. In healthy individuals, cells of the immune system allow us to live in symbiosis with micro-organisms, however the immune system is in disbalance in various diseases such as infectious diseases, cancer and auto-immunity.

The course will give the student the opportunity to enhance the knowledge on the scientific aspects within the field of immunology. Special focus lies on the immunological processes underlying homeostasis control i.e., tolerance induction, immunity, antigen presentation and processes that lead to the development of inflammatory diseases (infectious diseases through pathogens), autoimmunity (neuroimmunology) and cancer. Because this is an advanced course in the field of immunology, and will go into depth, particular on molecular details, students should be familiar with basic immunology preferably via a previous basic training course in immunology.

Additional Information Teaching Methods

The course covers immunological processes at the molecular level, and consists of lectures (22 hours) and study groups (14 hours). In the study groups students will read review articles as well as primary scientific articles on the different topics and discuss in groups opposing views on the molecular immunological processes that occur in the different stages of homeostasis and disease control. State of the art will be discussed of all topics, which will facilitate the study of scientific articles.
The course includes lectures, study groups, self-study, and in the the last part the course mainly covers a written assignment, self study and the exam. Contact hours with teachers and/or coordinators: 40

Method of Assessment

Study groups and active participation are compulsory. A substitution assignment is required when one or more workshops have not been attended, or when participation is judged as unsatisfactory.

A written exam (T) at the end of the course includes assay (‘open’) (90% of grade). A minimum score of 5.50 for the written exam is required in order to pass. Additionally, an assignment dedicated to scientific writing accounts for 10% of the grade.

Entry Requirements

Solid knowledge on basic immunology is compulsory before the start of the course.

Literature

Lectures, reviews and scientific papers are part of the material that covers the exam (posted on Canvas).

Additional Information Target Audience

Students with a keen interest to study immunological processes that form a basis for a variety of occurrences of diseases. In particular those that cover the interaction between host-pathogen, host–tumor and homeostatic control.

This is a compulsory course for MSc Biomedical Sciences students with specialization Immunology and/or Infectious Diseases.

AI in het Onderwijs

Course Code	O_MAIIHO
Credits	3
Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. ir. F.F.J. Hermans
Examiner	dr. H.B. Westbroek
Teaching Staff	prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

Het doel van dit vak is op basis van kennis over AI en over onderwijs theorieën een helder beeld schetsen van hoe

je je als docent en school wilt verhouden tot generatieve AG.

Course Content

Studenten leren in dit vak:

- Hoe generatieve AI werkt
- Wat de impact van AI op onderwijs kan zijn, vanuit een aantal verschillende theorieën en invalshoeken
- Wat de risico's van AI zijn, bijvoorbeeld mbt bias en deskilling
- Reflecteren op het gebruik van AI in de eigen onderwijspraktijk

Belangrijk: Voor dit vak is geen technische voorkennis vereist en het is voor alle studenten van de lerarenopleiding (en aanverwante studies) geschikt, zeker niet louter voor de beta's!

Additional Information Teaching Methods

Het vak bestaat uit:

- 5 verplichte hoorcolleges (10 uur)
- Huiswerkopdrachten maken (25 uur)
- Lezen literatuur (25 uur)
- Uitwerken beleidsplan (24 uur)

Method of Assessment

Studenten sluiten dit vak af door het schrijven van een beleidsplan over het gebruik van AI binnen hun vak of school.

Literature

Zie Canvas

Analysis of Governmental Policy

Course Code	AM_470571
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K.P.W. Kok
Examiner	dr. K.P.W. Kok
Teaching Staff	dr. K.P.W. Kok, K.E. Mujica Gastelbondo MSc
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

- To acquire critical knowledge on, and to master the correct use of, different policy models and theories;
- To further deepen your analytic skills with respect to the critical assessment of a complex societal question or dilemma in the health and life sciences;
- To learn to integrate scientific and non-scientific knowledge and skills;
- To practice skills in data collection and analysis;
- To learn to translate research findings into policy recommendations;
- To get experienced in writing a policy advisory report and to set up valid lines of argumentation;
- To improve your (communication) skills in working effectively in a project team through team building, team analysis and feedback.

Course Content

Governmental policy affects millions of people and is thus object of intensive debate and target of strong societal forces, including those exercised by political parties, media and interest groups. Being a policy advisor or policymaker requires a thorough understanding of the dynamics of policymaking, relating to (scientific) knowledge and expertise, interests and needs of individuals and organizations, as well as broader social structures and their influence on (deliberative) democracy.

The course contains several lectures on theoretical concepts and models concerning analysis of policy and analysis for policymaking. Furthermore you will be challenged, under supervision, to apply and practice these concepts and models in a problem-based real-world research project with transdisciplinary elements. From the very first day, you will be part of a project team of about ten students. You are confronted with a real policy problem from an external commissioning institution (e. g. a non-governmental organization, a Ministry, an advisory council). During the course you will collect data by literature review and interviews and conduct an interdisciplinary analysis on the basis of which you provide an advice, written down in a report. Specific attention is paid to working in a project team and team building. On the last day of the course you present the outcomes of your report to a representative of the external institution who commissioned the project. In that presentation your team will highlight the main results of your analysis and defend the recommendations you propose.

Additional Information Teaching Methods

Analysis of Governmental Policy is a part-time course of eight weeks (6 ECTS). Tuition methods include lectures, training workshops, and self-study. Attendance to lectures and project meetings is compulsory. In our experience, relying on self-study alone is insufficient to pass the exam.

Method of Assessment

40% Exam
40% Project report
10% Individual project performance
10% Presentation

Literature

- Buse, K., et al. (2023). Making health policy. 3rd edition. McGraw-Hill Education (UK). Note that this year we use the 3rd edition of this book.

Additional Information Target Audience

Compulsory course within the Masterprogramme Management, Policy Analysis and Entrepreneurship for the health and life sciences (MPA) and the Societal differentiation of Health, Life and Natural Sciences Masters programmes. The course can be followed as an optional course for other Science Faculty Master program students at the VU University, but only when taken in parallel with Research Methods for Analyzing Complex Problems AM_1182). Please contact the course coordinator prior to enrollment if you belong to this group.

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course, based on the actual number of students that start with the
course. It is not possible to enrol for workgroups otherwise - unless announced via Canvas.

Recommended background knowledge

The project integrates the research design made and lessons learned from the compulsory MPA course "Research Methods for Analyzing Complex Problems."

Business Management in Health and Life Sciences

Course Code	AM_470584
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K.D.S. Fernald MSc
Examiner	dr. K.D.S. Fernald MSc
Teaching Staff	dr. L.H.M. van de Burgwal, dr. E.M. Janse, dr. K.D.S. Fernald MSc
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

1. To acquire knowledge and understanding into theory of knowledge valorisation in health and life sciences

2. To acquire knowledge and insight in how to organise, protect and finance a business in health and life sciences
3. To acquire knowledge and understanding into the pharmaceutical industry's business model and business processes
4. To acquire knowledge and understanding into the challenges that the pharmaceutical industry faces
5. To apply newly acquired knowledge and understanding by solving case examples
6. To apply newly acquired knowledge and understanding in writing a business plan
7. To reflect on and critically evaluate the role of the pharmaceutical industry in the healthcare system
8. To learn to autonomously write a (personal) business plan

Course Content

As a result of external factors (for example ageing of the population and technological advancement, leading to increased healthcare costs), our healthcare system is under pressure. As a central stakeholder in this healthcare system, the pharmaceutical industry is facing significant challenges. Business Management in the Health and Life Sciences focuses on gaining insight in the pharmaceutical industry, its business model, business processes challenges, as well as strategies and actions to overcome these challenges.

During the course, dr. Linda van de Burwal and dr. Kenneth Fernald, together with highly experienced guest lecturers from the field will teach theoretical and practical knowledge during lectures and seminars.
Tangible subjects that will be discussed during the lectures and seminars include the pharmaceutical industry's business model and business processes, intellectual property, portfolio management, finance, risk capital, grants and subsidies, team building and people management, different legal entities, fiscal and legal aspects when starting a new company, SWOT analysis in the life sciences and clinical trials.

The newly acquired knowledge is tested via an assignment during which students will write either a personal career business plan or a 'real' business plan (50% of the final grade), a financial plan for the personal BP (30% of the final grade), patent seminars group assignment (15% of the final grade), and a final reflection video (5% of the final grade).

Additional Information Teaching Methods

Lectures: +-50 h
Computer seminars: 7,5 h
Work on assignment and self-study: +- 40h

Method of Assessment

- elevator pitch dream : pass/ fail
- providing feedback on pitch : pass/fail
- personal career business plan : 50 % of the final grade
- financial plan for the personal BP : 30 % of the final grade
- patent seminars group assignment : 15 % of the final grade
- final reflection video : 5 % of the final gradea
- final online exam : pass or fail

All parts have to be passed successfully.

Literature

Lectures provide key material to this course. The following book can be used as background to the lectures:

- Osterwalder, A. & Pigneur, Y. (2009). Business model generation. Self-published.

The course's assignment is to write either a 'real' business plan or a 'personal' business plan. When writing a business plan, students are recommended to use the following books:

- Kubr, Marchesi & Ilar (McKinsey & company). (1998). Starting up. Achieving success with professional business planning. McKinsey & Company, Inc. The Netherlands, Amstel 344, 1017 AS Amsterdam.
- Clark (2012). Business Model You: A One-Page Method for Reinventing Your Career. Business Model You, LLC.

Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences.

Additional Information

Guest lecturers contributing to this course include*:

- Olaf Larsen, Yakult
- Bart van Wezenbeek, V.O. Patents & Trademarks
- Maurice Mooijman, RVO NL Octrooicentrum
- Marga Janse, InnovatiefLerenLeren
- Robert Al, TU/e Innovation Lab
- Loyens Loeff Belastingadviseurs
- Bart Bergstein, Forbion Capital partners
- IQVIA
- Hans Slobbe, TalentMark

Clinical Development and Clinical Trials

Course Code	AM_1180
Credits	3
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A.M. van der Geest
Examiner	dr. A.M. van der Geest
Teaching Staff	dr. A.M. van der Geest
Teaching method(s)	Study Group, Lecture

Course Objective

- To gain knowledge and insight into the function clinical trials in today's healthcare system.
- To gain knowledge and insight into the design of clinical trials.
- To gain knowledge and insight into the conduct of clinical trials, including the applying rules and regulations (including ICH-GCP)
- To gain knowledge and insight into and critically reflect on the roles, tasks and responsibilities of the stakeholders involved in clinical trials
- To gain insight into challenges in clinical development as well as in strategies to deal with these challenges
- To learn where and how to look up rules and regulations

Course Content

In today’s healthcare system, clinical trials have gained the status of golden standard to test the safety and efficacy of newly developed drugs. For new drugs to enter the market, clinical trials must be passed and as a consequence, clinical trial outcomes have major effects on our healthcare system. While our healthcare system currently is under pressure to remain affordable and available to all, at the same time, clinical trial regulations are increasingly tightened and the prominence of clinical trials in our healthcare system is being criticized. For that matter, it is of great importance to learn about and reflect on the role of clinical trials in today’s healthcare system.

The Clinical Development & Clinical Trials course will elaborate on the function, design and conduct of clinical trials, as well as the relevant stakeholders involved. The course consists of a theoretical part and an important practical part (e.g. gaining knowledge on clinical trial regulations). Classes include for example: ‘Life Cycle of a Clinical Trial’, ‘Clinical Trial Methodology’, ‘ICH-GCP Principles’, ‘The Ethics Committee’, ‘Safety Considerations in Clinical Trials’, ‘Quality Control & Quality Assurance’, ‘Compliance, Misconduct & Fraud’.

The gained knowledge and skills will be evaluated by means of a two individual assignments, one group assignment, and an exam at the end of the course.

Additional Information Teaching Methods

Lectures: +-35 h
Self study: +- 40 h

Method of Assessment

Assessment;
Two individual assignments (each 20%), one group assignment (20%), and an exam (40%) at the end of the course.
All separate assignments must be sufficient (>5,5).

Literature

- Ray, S., Fitzpatrick, S., Golubic, R. & Fisher, S. (2016). Oxford Handbook of Clinical and Healthcare Research. Oxford University Press,Oxford, UK.
- (Additional reading will be provided via Canvas and will serve as background reading for the lectures).

Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences.

Additional Information

Guest lecturers, organizations/ companies:
• Eric Klaver, Trial&Eric.com/ IQVIA
* Guest lecture: SEUSS + working in the CT field.

Clinical Immunology and Disease Mechanisms

Course Code	AM_1309
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	T. Rispens
Examiner	T. Rispens
Teaching Staff	dr. G. Kooij, T. Rispens, prof. dr. R.E. Mebius, dr. J. van Horssen, dr. ing. S.J. van Vliet
Teaching method(s)	Seminar, Lecture

Course Objective

Course goal and learning objectives

- To understand immunopathogenic processes that play a role in the onset and chronicity of immunological diseases, that cover allergy, auto-immunity and infection diseases, such as celiac disease, inflammatory bowel disease, systemic lupus erythematosus, multiple sclerosis and HIV/AIDS.
- To acquire insight in both clinical parameters as well as basic scientific principles that play a role in these diseases.
- To acquire insights in the currently used treatments to reduce disease activity.
- To understand the mechanism by which the immune system regulates these diseases, and how one could modify immune response to the benefit of the patient.
- To apply the acquired knowledge on applicable clinical situations, like transplantation immunology (including graft rejection and graft versus host reactions), with a focus on immunopathology.
- To apply the acquired knowledge of scientific literature and scientific hypotheses of each of the topics described above in a written report as well as in debating clubs where disease-related statements will be defended
- To critically evaluate a scientific manuscript and provide a review report.

Course Content

Course content

During the course, various immunological diseases will be discussed one-by-one with a focus on multiple sclerosis, celiac disease and SLE and primary/secondary immunodeficiency's. Each theme will start with a clinical introduction into the features of the disease by a practicing clinician at the Amsterdam UMC, who illustrates the symptoms in patients that have these diseases. During each period a certain number of lectures will be given on the immunological mechanisms that play a role during these complex diseases.

These lectures highlight molecular immunological tools used, as well as novel strategies such as genomics-proteomics profiling of the disease, the use of animal models that mimic disease, as well as vaccine development and treatment methodology of the diseases. In turn, the students are able to apply the obtained knowledge in debating clubs, where disease-related statements have to be defended. These debating clubs are scheduled at the end of each theme. Moreover, in week 7, the students will hand in an individually written report (manuscript review)

to critically evaluate a scientific publication. All information on this report and the debating clubs (content and assessment plan) will be posted on Canvas.

Additional Information Teaching Methods

The course covers immunological processes as well as clinical parameters both at the molecular as well as the cellular level and will discuss both innate and adaptive immune responses. The course consists of lectures, self-study, a written report and debating clubs. Diseases are discussed one-by-one and the final part of the course mainly covers self-study and the exam.

Method of Assessment

The exam contributes for 75% to the result, the report for 10% and the debating club participation 15% to the result of your exam. For the exam, report and debating club a 5.50 score is required in order to pass.

Entry Requirements

Requested knowledge and literature

Bachelor’s course Immunology. A solid base on knowledge on immunology is compulsory before start of the course.

Immunobiology by Janeway 8th edition or Parham 4th edition, including a handout which contains recent reviews specialized on the immunological diseases discussed. Next to that, self-study and literature search is required to be prepared for debating clubs.

Literature

see canvas

Additional Information Target Audience

Additional information target group

Students with a keen interest to study immunological processes within the complexity of diseases such as allergy, multiple sclerosis and HIV/AIDS. Given the broad relevance of immunology in life sciences this course also provides excellent opportunities for other differentiations and Master programs.

Compulsory course for MSc Biomedical Sciences students with specialization Immunology.

Students who have followed the minor Cancer-Immunity-Personalized therapies (or the previous Immunity-Infection-Cancer Connected) can't participate in this course because of the overlap.

Clinical neurosciences

Course Code	AM_1005
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Donega
Examiner	V. Donega
Teaching Staff	dr. C. Vriend, V. Donega, dr. D.P. Bakker
Teaching method(s)	Practical, Written partial exam, Lecture

Course Objective

The aim of this course is twofold:

1. Provide knowledge on the clinical features, pathophysiology, molecular and cellular mechanisms, and latest developments in therapeutic strategies for a select number of neurological and neuropsychiatric disorders.
2. Initiate a translational way of thinking as a neuroscientist.

This course will provide the latest clinical and neurobiological knowledge on a number of key brain disorders:

Multiple Sclerosis, Childhood White Matter Disorders, Parkinson's disease, Neuropsychiatric Disorders, and Glioma.

It also introduces some of the tools that researchers have at their disposal to investigate the pathophysiology of brain disorders in vivo, such as neuroimaging.

This course will stimulate translational thinking by continuously crosslinking clinical features, (histo)pathology, and imaging abnormalities for diagnosis, prognosis, and therapeutic strategies.

Given the diversity of subjects and to provide the most up-to-date insight into the pathophysiology of the brain disorders, lectures are provided by both (external) pre-clinical and clinical researchers that are at the top of their respective fields.

Course Content

The course evolves around the following brain disorders and diseases Multiple Sclerosis, Childhood White Matter Disorders, Parkinson's disease, Neuropsychiatric Disorders, and Glioma.

Multiple sclerosis:
Description of the clinical phenotypes of the disease. Introduction of neuroimaging specifically adapted for MS. The use of neuropathology to understand disease mechanisms in MS.

Childhood white matter disorders (CWMD):
Introduction to the main clinical manifestations and different forms of CWMD. It will provide an example of translational research: from patient to gene, protein, and treatment.

Movement disorders:
The focus will be on Parkinson's disease. Description of the motor and cognitive manifestations of movements Disorders, and their underlying neurobiology. The latest research on the underlying molecular mechanisms, and therapeutic strategies will be discussed.

Neuropsychiatric disorders:
Description of the clinical manifestation of some of the main neuropsychiatric disorders, such as obsessive-compulsive disorder, major depressive disorder, and post-traumatic stress disorder. Introduction to the underlying neuroanatomy and pathophysiology and the role of neuroimaging in neuropsychiatry. The latest therapeutic treatments will be discussed.

Neuro-oncology:
Description of the clinical/cognitive manifestations of brain tumors. The focus will be on glioma. The effects of these tumors on brain networks is the main focus.

The course will additionally assess the students' skills to:
Summarize and present the essence of a scientific paper in the form of a graphical abstract and an oral presentation. Students are expected to participate actively during these assignments.

Additional Information Teaching Methods

Lectures, student presentations, and group assignments (live on campus). The course runs for eight weeks, two days per week.

Method of Assessment

- Group assignment to produce a graphical abstract on a recent cutting-edge paper in the field of Clinical Neuroscience (10% of grade);
- Group presentation on recent cutting-edge paper in the field of Clinical Neuroscience (30% of grade);
- A written digital exam with (primarily) open questions about all the course material (60% grade).

Sufficient grades (>5.5) for the presentation, as well as the exam are required in order to pass the course. Individual grades cannot be averaged to obtain a sufficient grade.

Entry Requirements

All students following the Neurosciences masters program at the VU are required to take this course.

This course is not open for external candidates.

Literature

Course slides that will be provided (exam material) and research papers (assignment material).

Additional Information Target Audience

MSc Neuroscience students;

Additional Information

Questions? clinicalneuroscience@amsterdamumc.nl

This course is coordinated by: dr. V. Donega (Vanessa) and dr. D.P. Bakker (Dewi)

Custom Course Registration

Enrolment through vu-net.

Recommended background knowledge

Knowledge on neuroanatomy, (basic knowledge) brain disorders, and basic knowledge on cell biology is**highly** recommended. Recommended literature: Neuroanatomy: an illustrated colour text (5th edition, Crossman & Neary, ISBN 9780702054051) and Molecular Biology of the Cell (Alberts et al., ISBN-10: 0-8153-4072-9).

Communication, Organization and Management

Course Code	AM_470572
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C.A.C.M. Pittens MSc
Examiner	dr. C.A.C.M. Pittens MSc
Teaching Staff	dr. S.S. Rai, dr. R.M.H. Peters, dr. C.A.C.M. Pittens MSc
Teaching method(s)	Study Group, Lecture

Course Objective

After completion of this course, you will be able to:

- identify and discuss knowledge of, and theories on organisational structures, organisational behavior, organisational change, and change management;
- describe and conceptualise organisations as embedded in a wider system, and interpret these from a transdisciplinary perspective;
- obtain a deeper understanding of communication from the perspective of sharing and influencing results, and to learn how to deal with conflicts;
- identify and discuss theories on leadership and interpersonal behaviour;
- make an informed choice between statistical techniques commonly used in health and life sciences, applying these to conduct analyses while using statistical software (e.g., SPSS) and interpreting the results of these techniques (e.g., t-tests, chi-square tests) to inform decision making;
- practise analytical and advisory skills.

Course Content

Organisations need to respond and adapt to internal and external challenges. It is increasingly recognized that addressing these challenges is not an ‘isolated process’. Organisations (need to) collaborate with multiple stakeholders in these processes of (continuous) change. In-depth understanding of the structure of an organisation and the wider system it operates in is therefore important. Different forms of knowledge can be used to gain this understanding. It requires insights gained from the analyses of qualitative and quantitative data collected within organisations and their wider context. This course offers insights into these complexities, as well as practical (quantitative) methods to address these.

In interactive lectures, several (guest)speakers will discuss macro-aspects of organisational behavior, including designing organisations and managing them, thereby touching on the managerial skills and the strategic thinking required. In addition, concepts related to motivation, managing interpersonal behaviour, leadership, communication and developing and changing organisations will be explained. Furthermore, lecturers will discuss concepts of systems thinking, and the value of adopting multiple perspectives in analysing organisations and processes of organisational change. The (guest) speakers will illuminate concepts derived from the literature by relating them to

their own experiences with informing and managing organisational change in practice. Furthermore, the role of quantitative data in managerial decision making will be explored (the role of qualitative data is covered in other courses). To that end, you will explore and discuss different statistical techniques (e.g., t-tests, chi-square tests) commonly used in health and life sciences. You will conduct analyses with these techniques in statistical software (e.g., SPSS), interpret the results and translate these to decision making processes, working towards an advisory report for a (fictive or real-world) organisation's management.

In an assignment that runs through the course, practical cases of healthcare companies will be analysed and discussed, resulting in advisory reports for management. With the other students you discuss your experiences and a coach helps you relate the experiences to theory.

Additional Information Teaching Methods

Lectures: approximately 22 hours
Response lectures: 4 hours
Training workshops: 16 hours (attendance is mandatory)
Self-study and writing project assignment: remaining hours.

Method of Assessment

Written exam (60%) and assignment (40%). Grades of both components must be sufficient.

Literature

To be announced on Canvas

Additional Information Target Audience

Compulsory course within the Master programme Management, Policy Analysis and Entrepreneurship for the Health and Life Sciences (MPA) and the Societal differentiation of Health, Life and Natural Sciences Masters programmes

Additional Information

Attendance to training workshops / working groups is indispensable

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course. It is not possible to enrol for workgroups otherwise unless announced via Canvas.

Containment Strategies of Infectious Diseases in Global Context

Course Code	AM_470127
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.R. Essink
Examiner	dr. D.R. Essink
Teaching Staff	dr. D.R. Essink
Teaching method(s)	Study Group, Lecture

Course Objective

The endpoint of this course is that the student

- Has acquired in-depth theoretical and practical knowledge in relation to health intervention strategies for infectious diseases.
- Has acquired insights in various infectious diseases and characteristics in relation to containment strategies.
- Has acquired insight into the role of international institutions, such as the WHO, governmental advisory bodies, relevant professionals, executing institutions, NGOs and communities in designing and carrying out

health interventions.
- Understands which barriers are important when implementing containment strategies of infectious diseases, with a focus on vaccination programs.
- Has acquired insight in theoretical concepts and methods to interpret results, evaluations and the effectiveness of programs.
- Has learned and practiced interdisciplinary methods and techniques to plan health interventions at community level in an interactive way.
- Is able to communicate results by means of an evidence based policy brief, a video and orally

Course Content

The COVID-19 pandemic is clear evidence that this course has been and still is relevant. This course focuses on the design and implementation of intervention strategies used to address infectious diseases in a global context. The approach taught to
design interventions is also applicable beyond the domain of infectious disease control.
The containment strategies to be discussed include programmes for known infections (including vaccination strategies and
in case of absence of a vaccine, diagnosis and treatment strategies, environmental and social control strategies) and emerging infections (including isolation, prevention and communication strategies).
The student learns how to analyze bottlenecks and opportunities of the various strategies, how to interpret the results and to evaluate the implementation of programmes.
In addition, the student will take part in a group assignment on how to design containment strategies at community level in an interactive way, for e.g. tuberculosis, polio, rabies, malaria, HIV/AIDS, Ebola, etc. A video presentation and writing of a policy brief will be part of the group assignment.

Additional Information Teaching Methods

Lectures, group assigment, (video) presentation, essay, self-study. Basic background knowledge will be provided by VU lecturers, whereas relevant guest lecturers will present practical field examples.

Group assignment attendance is compulsory.

Contact hours: lectures 34 hrs, group work 8 hrs.
Self-study approx. 80 hrs.

Method of Assessment

Individual exam (60%) and group assignment presentation and policy brief (40%).
Both parts must at least be sufficient (6 or higher)

Entry Requirements

Basic knowledge about the pathogenesis of infectious diseases, microbiology and immunology

Literature

BOOK: Webber, Roger. Communicable diseases: a global perspective. Cabi, 2019.
or
R. Webber, 2016. Communicable diseases: a global perspective. ISBN 9781780647425 / DOI 10.1079/9781780647425.0000

Lecture Slides: Slide of lectures as made available on Canvas.

Additional readings: Lecturers may make further readings available on Canvas.

Additional Information Target Audience

Compulsory course within the Master differentiation International Public Health; optional course for students in other differentiations of the Masters Health Sciences, Biomedical Sciences, and Management, Policy Analysis and Entrepreneurship in Health and Life Sciences. Students from other backgrounds, please contact our secretariat for further information at secretariaat.athena@vu.nl

Additional Information

VU lecturers:
Prof. dr. Aura Timen
Dr. Dirk Essink

Guest lecturers:

Dr. M. Gidado (KNCV)
Dr. Constance Schultsz (AIGHD/AMC)
Prof. dr. Maarten Postma (RUG)
Dr. Koert Ritmeijer (MSF)
Prof. dr. Cees Hamelink (VU)
Prof. dr. Ron Fouchier (EMC Rotterdam)
Dr. Hans Zaaijer (Sanguin)
Aula Abara, Msc. (MSF)
and others

Custom Course Registration

Enrollment through Canvas.

Didactics 1

Course Code	O_MLDIDAC_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	prof. dr. M. Meeter, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, drs. M.A. Huijskes MA, dr. H.B. Westbroek, dr. C.D.P. van Oeveren, drs. A.J.C. Monquil, dr. O.P.M. van Buuren, drs. Y.G. Meindersma, C.L. Geraedts, dr. B. de Vries, drs. R. Cornelisse MSc, M. Beukering, drs. E.D. van Noort, drs. H.R. Goudsmit, drs. M.M. Bout, prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, drs. J.K.W. Riksen, drs. N.H. Ypenburg
Teaching method(s)	Seminar, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio's vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben

doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactics 1

Course Code	O_MLDIDAC_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts

Teaching Staff	drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, drs. J.K.W. Riksen, dr. O.P.M. van Buuren, drs. E.D. van Noort, drs. Y.G. Meindersma, dr. S.M. Calor, dr. I. Pauw, drs. A.J.C. Monquil, M. Beukering, drs. N.H. Ypenburg, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. M.M. Bout, drs. H.R. Goudsmit, dr. H.B. Westbroek, C.L. Geraedts, N. el Abdellaoui MSc, drs. R. Cornelisse MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio’s vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio’s werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Interactieve hoorcolleges, werkcolleges en zelfstudiemodules.

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactiek 2 Semester 1

Course Code	O_MLDIDAC2
Credits	9
Period	P2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	C.L. Geraedts, prof. dr. M. Meeter, drs. M.M. Bout, drs. A.J.C. Monquil, drs. E.D. van Noort, drs. R. Cornelisse MSc, dr. H.B. Westbroek, dr. S. Donszelmann, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, N. el Abdellaoui MSc, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, drs. L.J. van Well-van Grootheest, drs. H.R. Goudsmit, drs. Y.G. Meindersma, M. Beukering, drs. M.A. Huijskes MA, drs. N.H. Ypenburg, dr. I. Pauw, dr. S.M. Calor
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een

peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema's centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

NB: Voor studenten die vóór 1 september 2023 gestart zijn geldt in studiejaar 2023/2024 het curriculum van 2022/2023. Dit betekent dat het vak Didactiek 2 een studielast van 6 EC (ipv 9EC) heeft (en ze zich voor die variant intekenen). Studenten die de colleges van Didactiek 2 (6 EC) nog niet (geheel) gevolgd hebben sluiten aan bij colleges van het nieuwe vak Didactiek 2 (9 EC). Het is tot en met augustus 2024 nog mogelijk om Didactiek 2 (6EC) uit het oude curriculum af te ronden en het examen onder de eindtermen van het oude curriculum af te leggen.

Didactiek 2 Semester 2

Course Code	O_MLDIDAC2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. Y.G. Meindersma, drs. E.D. van Noort, prof. dr. ir. F.F.J. Hermans, drs. N.H. Ypenburg, dr. O.P.M. van Buuren, dr. H.B. Westbroek, dr. I. Pauw, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, dr. B. de Vries, drs. A.J.C. Monquil, M. Beukering, C.L. Geraedts, drs. R. Cornelisse MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, drs. M.A. Huijskes MA, D.S. Hoefakker-Laanstra, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. M.M. Bout, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio’s werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema’s centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen

met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

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Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

NB: Voor studenten die vóór 1 september 2023 gestart zijn geldt in studiejaar 2023/2024 het curriculum van 2022/2023. Dit betekent dat het vak Didactiek 2 een studielast van 6 EC (ipv 9EC) heeft (en ze zich voor die variant intekenen). Studenten die de colleges van Didactiek 2 (6 EC) nog niet (geheel) gevolgd hebben sluiten aan bij colleges van het nieuwe vak Didactiek 2 (9 EC). Het is tot en met augustus 2024 nog mogelijk om Didactiek 2 (6EC) uit het oude curriculum af te ronden en het examen onder de eindtermen van het oude curriculum af te leggen.

Disability and Development

Course Code	AM_470588
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.M.H. Peters

Examiner	dr. R.M.H. Peters
Teaching Staff	dr. R.M.H. Peters
Teaching method(s)	Study Group, Lecture

Course Objective

- To define the concepts of 'impairment', ‘disability’ and ‘inclusive development’ using examples
- To explain the complex two-way relation between disability and poverty
- To explain the issues and barriers faced by people with disabilities in the global context
- To demonstrate with examples the importance of the UN Convention for the Rights of Persons with Disabilities and the SDG Framework for the field of disability and development
- To apply and compare relevant theories, models and frameworks related to disability (e.g. medical model, social model, ICF model, ecological system theory, twin-track approach, stigma, intersectionality, epistemic injustice).
- To design and assess studies related to disability (including participatory approaches) and interventions.
- To apply research skills during the problem-based learning sessions (e.g., participating in scientific discussions, formulating of research objectives, literature research, abstracting, summarizing and giving feedback on findings, drawing conclusions)

Course Content

The Disability and Development (D&D) course focuses on a broad range of issues related to disability in the global context. Disability affects an estimated 1.3 billion people worldwide. With trends such as aging, medical advances, population growth, shifting disease burden and the climate crises these numbers are expected to increase. About 80% of the persons with disabilities live in low and middle income countries. In countries with life expectancies over 70 years, individuals spend on average 8 years living with disabilities. Across the globe, a substantial number of people with disabilities have limited access to education, employment and health (rehabilitation) services. They also may have to deal with stigmatization, social exclusion and discrimination.

The United Nations Convention to the Rights of People with Disabilities (UNCRPD) has been crucial for the promotion of disability-inclusive development. This treaty was adopted in 2006 and has 164 signatories as of 2022. Parties that signed the convention are required to promote, protect, and ensure the full enjoyment of human rights by persons with disabilities. They also need to ensure that people with disabilities enjoy full equality under the law. Making the goals set out by the UNCRPD a reality is work in progress. Also the Sustainable Development Goals (SDG) framework, with its commitment “to leave no one behind” has been crucial. The framework explicitly emphasizes including people with disabilities in terms of education, employment and research, reducing inequality, and creating inclusive communities. In 2019, heads of state and government representatives adopted the political declaration: “Universal health coverage: moving together to build a healthier world”. This declaration includes a specific reference made to persons with disabilities. In 2021 a landmark resolution was adopted by the World Health Assembly, on “The highest attainable standard of health for persons with disabilities”, which aims to advance the agenda of disability inclusion in the health sector in countries, and focuses on three central areas: i) access to effective health services; ii) protection during health emergencies; and iii) access to cross-sectorial public health interventions.

This course will cover essential knowledge and skills in the field of Disability in the Global Context.

Students are encouraged to think critically, ethically, holistically, rationally, and collegially about topics addressed in this 4 week course programme, such as:

- Epistemic Injustice
- Disability Models and other relevant theories
- Intersectionality
- Global Mental Health
- Disability and Sexuality
- Role of families
- Gender
- Violence
- Researching Disability and Stigma
- Disability Rights
- Activism
- Migration
- Child exploitation and the NGO perspective

Additional Information Teaching Methods

Problem-based learning supported by lectures.

- Lectures: 21 hours

- Tutorial groups: 21 hours
- A few optional oral exam preparation sessions
- Self-study + portfolio : remaining hours

Method of Assessment

- Oral examination (60%)
- Portfolio (30%)
- Participation in the tutorial group work (10%)
- Peer feedback (sufficient/ insufficient)

For all parts, a passing grade (> 5.5) needs to be obtained in order to successfully finish the course.

Entry Requirements

Bachelor-level education; any subject

Literature

See Canvas for suggested readings

Additional Information Target Audience

The Disability & Development module is an optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Health Sciences (HS), Global Health (GH) and Biomedical Sciences (BMS).

Additional Information

For more information contact Dr. Ruth Peters (r.m.h.peters@vu.nl)

Epidemiology

Course Code	AM_1179
Credits	3
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.S. Rai
Examiner	dr. S.S. Rai
Teaching Staff	L.L. Ruijsch, K.E. Mujica Gastelbondo MSc, L.L. Kaersenhout MSc
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

- To be able to describe the key characteristics, strengths and weaknesses of traditional epidemiological study designs and select an appropriate study design for a given research question and context;
- To be able to understand, calculate, and apply measures of occurrence and association;
- To be able to understand and assess possible bias and effect modification;
- To be able to understand and apply principles of accuracy in epidemiology;
- To gain an understanding of the principles of screening and calculate related measures;
- To be able to critically interpret and evaluate epidemiological studies (choice of study design, analyses, results, possible biases, and practical ideas for improvement)

Course Content

This is a basic epidemiology course and consists of a theoretical, contextual, and practical component. The theoretical component will focus on concepts and methodology used in epidemiological studies (e.g. study-designs, measures of occurrence and association, screening, bias, etc.). The contextual component will focus on past and current epidemiological developments, for instance, the start of the HIV/AIDS pandemic. Lastly, the practical component will focus on application of all your new skills through workgroups and journal club (practical).

Note: *This course does not include biostatistics. This course will only focus on basic epidemiological concepts, indicators/measures and related calculations, and critical appraisal and evaluation of epidemiological studies.*

Additional Information Teaching Methods

- Lectures (14 hours)
- Work groups (12 hours)
- Journal club (8 hours)
- Assignment (8 hours)
- Self-study (remaining time)

Method of Assessment

- Exam (100%)
- Assignment (insufficient/ sufficient)

Both elements need to be sufficient.

Literature

Available on Canvas

Additional Information Target Audience

This course is solely intended for students without a background in epidemiology(i.e. students who attended and completed another bachelor or master course in public-health or social sciences related methodology and applied biostatistics, epidemiology and biostatistics, or similar, are strongly discouraged from enrolling in this course).

Additional Information

For more information contact Dr. Sarju Sing Rai (s.s.rai@vu.nl)

Custom Course Registration

n/a

Ethics in Life Sciences

Course Code	AM_470707
Credits	3
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	P.R. Collard MA MSc.
Examiner	P.R. Collard MA MSc.
Teaching Staff	dr. C. Boersma MA MSc., P.R. Collard MA MSc.
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

This course:

- Provides students with knowledge of central concepts and theories in applied philosophy and professional ethics;
- Provides students with a toolbox of ethical instruments to help analyze moral problems related to (one's own) research in the life sciences and beyond.

Students who have taken and successfully finished this course:

- Are able to reflect on ethical issues related to one’s own life science specialization;
- Are capable of impartial and constructive deliberation concerning ethical issues related to one’s own life science specialization;
- Can work in teams on (hypothetical) scientific or technological developments, taking into account multiple values (amongst which safety);
- Have acquired the necessary skills to handle ethical issues in an accountable manner, as a professional academic beyond one's own inclinations and prejudices;

- Have shown a respectful and accountable attitude in dealing with group dynamics during the seminar groups and group project.

Course Content

Researchers in life sciences help generate knowledge that builds the future of our society. Therefore, professional academics should be accountable for their decisions, experimental designs and presentation of results. In this course, the principles of justification will be illustrated with cases of technology ethics, medical ethics and care ethics. An ethical dialogue on animal research is simulated through a role-play exercise. Finally, as a group training project, an ethical deliberation on a developed product that is safe by its design is prepared and executed in confrontation with another team.

Additional Information Teaching Methods

Ethics in the Life Sciences is a part-time course of four weeks (3 ECTS).

The total study time is 80 hours.

The different elements have the following study time:

- Lectures: 13 hours
- Seminar groups: 15 hours
- Group assignment: 20 hours
- Exam: 2 hour
- Group project presentation and deliberation: 3 hours
- Self study: 27 hours

Please note that attendance to the seminars is compulsory. Attendance to the lectures is highly recommended. In our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the seminars, and to pass the exam.

Method of Assessment

- Intellectual participation in the seminars and group project (10%)
- Multiple-choice exam (50%)
- Execution of the group project, the presentation and deliberation (40%)

All three elements have to be passed.

Entry Requirements

Bsc Biology, Biomedical Sciences, Psychology with profile Biological Psychology or Neuropsychology

Literature

Available on Canvas

Additional Information Target Audience

Compulsory course in all former FALW Master programmes, except Health Sciences and Neuro Sciences

Additional Information

Lectures, seminars, the exam and the final assessment are in English.

From Molecule to Mind

Course Code	AM_1275
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	D.A. Maas
Examiner	prof. dr. H.D. Mansvelder

Teaching Staff	dr. G.J. Schenk, prof. dr. M. Verhage, prof. dr. H.D. Mansvelder, D.A. Maas, prof. dr. A.B. Smit
Teaching method(s)	Practical, Written partial exam, Lecture

Course Objective

Providing master students, independent of primary training, with a solid basis in molecular and cell biology, neurophysiology and functional neuroanatomy.

Course Content

The course will be used to brush up your knowledge of neurophysiology, molecular and cellular neurobiology and neuroanatomy. Lectures by established researchers on exciting and cutting-edge topics as well as a keynote symposium related to the different research programs on campus will be held to give you an impression of the research we do and to prepare you for your search for an internship project (which will start in February 2024).

Additional Information Teaching Methods

Lectures, study groups, assignments, practical sessions*, presentations, peer feedback, demonstrations.

*Please be aware the number of spots for the practicals is limited. In case the maximum number of students is exceeded, the course coordinator will decide who will be registered for the practicals, for the others there is an online practical available.

Method of Assessment

You will have to prepare presentations based on neuroanatomy related assignments and are expected to participate actively during practical sessions. You will receive a pass/fail grade for neuroanatomy. A 'pass' will be rewarded by a 30% correct starting grade for the final exam; an 'incomplete' will be rewarded by 0% (which means your maximum final grade for the course will be 70%). The final exam will consist of a digital multiple choice examination with two parts: cell biology and neurophysiology.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Literature

- Alberts et al., Essential Cell Biology, 4th edition (2013; ISBN 9780815344544). Purves et al., Neuroscience, 5th edition (2012; ISBN 9780878936953). NB. More recent print editions of these books may also be used.

Additional Information Target Audience

First year VU Master Neurosciences

Recommended background knowledge

We strongly advise you to study these chapters before the course starts.NB. Alternatively, study equivalent chapters from more recent printings of these books:

- Alberts et al., Essential Cell Biology, 4th edition (2013; ISBN 9780815344544). Chapters 4, 5, 7, 8, 15 - 17
- Purves et al., Neuroscience, 5th edition (2012; ISBN 9780878936953). Chapters: 3 - 8.

Genetics in Neuroscience

Course Code	AM_1214
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	J.E. Savage
Examiner	J.E. Savage
Teaching Staff	J.E. Savage

Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

After completing this course, students should be able to:

- Explain theoretical principles underlying human genetics
- Describe current state of the art methods for gene discovery for brain-related traits
- Apply computational methods to analyze large genetic datasets
- Compare tools that can be used to gain biological insight into genetic findings
- Critically evaluate results from genetic discovery studies of complex traits
- Reflect on how genetics and neuroscience can inform each other
- Apply genetic methods to inform the design of neurobiological studies

Course Content

The course is offered to students in the Master of Neurosciences. The main goal is to teach how statistical genetic studies are conducted to identify genes for complex neuropsychiatric traits and disorders. The course will discuss how results of such studies should be interpreted and how they can be used to inform neurobiological research. The lectures will provide a comprehensive overview of the current revolution in the field of human genetics, and will also lay out the theoretical foundations of this field. In addition, the lectures aim to provide starting points for bridging human genetics and neurobiology. Lectures will include examples of how to apply statistical and bioinformatic tools to gain insight into the genetics of brain-related traits. Practical sessions are aimed at providing hands-on experience with current analytical tools and fostering a critical opinion on published studies in this field.

Additional Information Teaching Methods

13 lectures (2 hours each), 10 computer practical sessions (3 hours each), and 1 presentation session.

Contact hours: 57 hours distributed over 24 sessions. Students are expected to read scientific papers to prepare for each lecture.

Lecture material: Lecture slides + scientific papers. Active participation in lectures and discussion sessions is expected.

Computer practical sessions consist of: - Unix programming - practical exercises using genetic tools (e.g. PLINK, FUMA, MAGMA, PRSice), covering: - quality control of genetic data - running a genome-wide association study - interpreting the results from a genome-wide association study in biological context - conducting genetic risk prediction based on the results from a genome-wide association study. Computer assignments need to be completed during the computer practical session or handed in via Canvas. Students will gain hands-on experience with the analysis pipeline used in genome-wide association studies.

Assignments and presentations will involve critical evaluation of scientific literature and application of statistical genetic methods to neuroscience research.

Literature: Scientific papers that will be provided throughout the course.

Method of Assessment

One final exam in week 8, one individual essay (graded) in week 4, one group presentation (graded) in week 7, and assignments for each computer practical. The final score is the weighted average of .65*exam + .10*essay + .20*presentation + .05*practicals. A sufficient grade (5.5. or higher) is required for both the exam and the overall score in order to pass the course.

Practical attendance is a necessary condition to pass this course. Practical sessions and presentations are mandatory.

Resit: Offered for the exam.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Literature

Scientific articles that will be provided throughout the course.

Additional Information Target Audience

First year VU Master Neurosciences students. This course is not open for external candidates.

Recommended background knowledge

Basic knowledge of genetics and/or cell biology, neuroscience or psychology, and familiarity with basic computer programming is recommended.

Health, Globalisation and Human Rights

Course Code	AM_470818
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.R. Essink
Examiner	dr. D.R. Essink
Teaching Staff	
Teaching method(s)	Study Group, Lecture

Course Objective

The student;

- Is able to describe, understand and apply human rights concepts in a global context
- Develops a deeper understanding and a critical attitude towards scientific literature in the field of health, globalization and human rights in order to formulate soundly argued positions
- Is able to create his/her own vision with regard to the socio-cultural dimensions of human rights values in relation to public health
- Is able to apply methods of human rights assessment in relation to innovations in health care
- Demonstrates the ability to write and present according to academic standards

Understand and apply scenario making in the context of human rights oriented global health futures

To understand and apply (research) ethics in global health

Course Content

This course focuses on the human rights issues that are raised around the globe in connection with global health concerns. The course introduces the students to the effects of globalization on health issues, to the relevant UN human rights instruments on health and to the mechanisms to promote and protect these rights. Attention is given to a wide range of human rights topics in which health and well being play a crucial role. Examples are situations of armed conflict, reproductive rights, migration and refugee issues and children's rights. Within the context of current globalization processes the importance of local cultural insights into the human rights & global health interaction will be discussed. During the course students will prepare and participate in a scenario/simulation exercise on a human rights assessment of innovations in health technology and discuss relevant scientific literature in study groups. During the course students consistently reflect on human rights and research ethics in the context of global health. In the exam students will show their creative problem-solving skills applying them to human rights dilemmas in public health.

Additional Information Teaching Methods

Contact hours

Lectures: 33 hours
Work groups: 12 hours
Group project, simulation and exam: 11 hours

Self study and preparing: remaining hours

Method of Assessment

Group projects (40%), exam (60%).
All parts need to be passed (6.0) (the exam is an essay written or oral)

Literature

Hamelink, C. J., Essink, D. R., & Visser, M. J. (Eds.). (2024). Global Health and Human Rights: Principles and Practices. Taylor & Francis.

To be announced at the start of the first work group/lecture and via Canvas for group project.

Additional Information Target Audience

Compulsory course for research master global health students. Optional course for students in all specializations of the Masters Health Sciences, Biomedical Sciences and Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences.

Additional Information

For more information contact Ona Ilozumba (ona.ilozumba@vu.nl)

Immune Modulation, Monitoring and Metabolism

Course Code	AM_1318
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.J. Garcia Vallejo MBA
Examiner	dr. J.J. Garcia Vallejo MBA
Teaching Staff	dr. G. Kooij, dr. J.J. Garcia Vallejo MBA, dr. ir. J. Van den Bossche, dr. S. Duinkerken, dr. ing. S.J. van Vliet, dr. J.M.M. den Haan
Teaching method(s)	Seminar, Study Group, Lecture

Course Objective

Drugs that **modulate** the immune system, including corticosteroids, antihistamines, vaccines, immune stimulants, and immunosuppressants, are widely used in treating immune-mediated disorders. Newer immune-modulating drugs, such as protein kinase inhibitors and biologics (including metabolite derivatives and metabolic drugs), are gaining attention across a range of human diseases. This rapid advancement in immunopharmacology, focusing on selectively regulating immune responses to meet clinical needs, is evolving into a distinct field.

With this modulation comes a need to **monitor** drug impact and identify patients who will benefit most from specific interventions. Given the immune system's complexity, single-cell technologies have emerged as key tools in developing immunotherapies and enabling detailed immune monitoring, enhancing our understanding of cellular behavior in immunity, infections, and cancer.

Immunometabolism, an emerging area exploring the interaction between metabolism and immune function, is also becoming pivotal for immune modulation and monitoring. By revealing the metabolic needs and weaknesses of immune cells, immunometabolism provides novel targets to influence immune responses and disease progression. It also introduces biomarkers for monitoring immune states, helping tailor treatments to individual immune and metabolic profiles.

This course aims to explore current and emerging immune intervention targets and monitoring methods for personalized medicine and therapeutic outcomes . Students will also write and review research proposals on immunological interventions and associated monitoring strategies, alongside expert discussions. Together, this course offers the latest insights in immune modulation, monitoring, and metabolism, while also building key academic skills for a scientific career.

Course Content

The course includes current and upcoming possibilities in immune modulation and the management of immune-mediated diseases. Different technologies to monitor immune responses and disease progression are covered, particularly focusing on next-generation single-cell and omics analyses. Approaches to modulate immune responses are discussed, encompassing immunotherapy in cancer, allergies, and autoimmune disorders. Also, since metabolism as an emerging regulator of immunity and disease, strategies to modulate and monitor immunometabolism are also addressed.

Additional Information Teaching Methods

Knowledge on immune modulation, monitoring and metabolism, obtained during lectures, self-study and meet-the-experts journal clubs, will be applied when writing a research proposal.

After an introduction and recap, the following key themes will be addressed:

- Current and new ways of immune modulation and immunotherapy
- Strategies and technologies to monitor the immune system and disease progression
- Metabolic aspects of immune modulation and monitoring

In addition to lectures, students will read two articles on the subject as self-study and will get a lecture of the main authors of the papers, followed by a meet-the-expert Q&A session.

The second part of the course consists of writing, reviewing and presenting a research project related to immune modulation, monitoring and metabolism in which the obtained knowledge is applied.

As such, in addition to learning how the immune system can be modulated and monitored for therapeutic gain, an important part of the course includes obtaining key academic skills such as critical reading, writing research proposals, reviewing and discussing articles.

Method of Assessment

Lectures, meet-the-expert sessions and research proposal presentations are compulsory, and form part of the material that covers the written exam at the end of week 8 (consisting of 50% of total points).

5% of the points is given for preparing the meet-the-expert sessions (reading articles and writing 50 words summary of it, listing strengths & weaknesses, propose follow-up research, provide questions for the expert author).

Writing and presenting a research proposal (including a hypothesis based on current literature and lectures, aims, design experiments, expected results) is an important part of the course and counts for 40% of the points (detailed below).

Reviewing each other's projects (complete peer review form) and asking questions during the presentations of the projects are important academic skills that count for 5% of the points.

Rubrics will be used to score the different writing assignments.

Written exam at the end of week 8 includes 'assay 'questions. A minimum of 5.50 for both the exam and practical parts (averaged) is required in order to pass.

Entry Requirements

Bachelor's courses on Immunology and Biochemistry.

Literature

Immunobiology by Janeway (8th edition) or Parham (4th edition) provides a solid basis for the immunological part of the course and Mark's Essentials of Medical Biochemistry provides background for the biochemistry aspect of the course. Immune modulation, monitoring and metabolism Immunometabolism are rapidly evolving and new fields for which no good and up-to-date books exist. Therefore, a handout will be provided that contains recent excellent reviews on the specific topics. Next to that, self-study and literature search is required to prepare for the meet-the-expert session and writing assignment. The review "A guide to immunometabolism for immunologists" provides a good first glimpse into this new and exciting field.

Additional Information Target Audience

Compulsory course for students MSc Biomedical Sciences - Specialization Immunology

The target group for this course are students with a keen interest to study metabolism and related immunological processes within the complexity of various diseases. Given the broad relevance of biochemistry and immunology in life sciences, this course also provides excellent opportunities for other differentiations and Master programs.

Inclusieve Digitale Didactiek

Course Code	O_MIDD

Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S.M. Calor
Examiner	dr. S.M. Calor
Teaching Staff	dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

De doordachte afstemming van complementaire face-to-face en online activiteiten moet duidelijk tot uiting komen in het ontwerp dat studenten leveren voor hun (nieuwe of bestaande cursus). Hiervoor gebruiken zij de VU Amsterdam Blended Learning (her) ontwerpsjabloon. Een belangrijk onderdeel hiervan betreft de motivatie die studenten gebruiken. Daarnaast lichten ze hun keuze voor online en face-to-face activiteiten toe en de volgorde hiervan, waarbij ze ook aandacht besteden aan constructive alignment; leerdoelen, leeractiviteiten en (formatieve en summatieve) testactiviteiten zijn op elkaar afgestemd. Inclusiviteit is een van de kernwaarden van de VU. Studenten gebruiken de VU Amsterdam Inclusiviteitskijkkader om de mate van inclusiviteit van het ontwerp in kaart te brengen. Ten slotte heeft de introductie van Generatieve Artificial Intelligence (AI) een enorm potentieel voor het onderwijs. Studenten gebruiken een VU Amsterdam doordachte didactiek bij de integratie van Generatieve AI in hun ontwerp.

Course Content

De coronaperiode heeft de inzet van digitale onderwijsvormen zoals online, blended en hybride onderwijs in een stroomversnelling gebracht. Naar aanleiding van het rapport Ruimte voor Onderwijs tijdens Corona en de motie Kwint is de wens ontstaan om meer inzicht te krijgen in de manier waarop digitale onderwijsvormen bijdragen aan het inhalen van coronavertragingen en het verbeteren van het welzijn van studenten. Daarnaast heeft de introductie van generatieve kunstmatige intelligentie (AI) een enorm potentieel voor het onderwijs. In het keuzevak “Inclusieve Digitale Didactiek” wordt aandacht besteed aan blended learning, inclusieve digitale didactiek en de integratie van generatieve AI in het onderwijs. Recente onderzoeksresultaten uit VU onderzoek worden in het keuzevak verwerkt.

De studenten van vandaag zijn opgegroeid in een wereld waar internet altijd aanwezig was, evenals mobiele telefoons en instant messaging. Ondersteunende leertechnologieën bieden een nieuwe manier om universitair onderwijs te ontwerpen, ontwikkelen en verzorgen en vormen een aanvulling op traditionele onderwijs- en leermethoden (Orton-Johnson, 2009). Als gevolg hiervan passen universiteiten en hogescholen hun financiering aan deze veelbelovende en nieuwe technologieën aan. De mogelijkheden voor communicatie, interactie en samenwerking worden uitgebreid, waardoor het hoger onderwijs naar het kantelpunt van een belangrijke transformatie wordt geduwd, waarvoor nieuwe vaardigheden van zowel studenten als docenten nodig zijn. Leertechnologieën kunnen ongekende onderwijsmogelijkheden bieden. Het vereist echter vakmanschap om effectieve blended learning, inclusieve lessen waarin generatieve AI in is geïntegreerd, te ontwerpen en ontwikkelen. Leraren moeten kennis hebben van inhoud, didactiek en technologie, evenals het vermogen om deze kennis zodanig toe te passen dat technologie het leren verbetert.

Additional Information Teaching Methods

Dit vak is opgebouwd uit een aantal colleges. Tussentijds bestuderen studenten relevante wetenschappelijke literatuur en werken zij aan hun onderwijs ontwerp voor een vak dat zij zelf geven. Tussendoor is er gelegenheid gebruik te maken van individuele coaching. Studenten presenteren hun tussentijds ontwerp aan elkaar en ontvangen van iedereen feedback. Deze peer feedback kan gebruikt worden ter voltooiing van het onderwijsontwerp. Tijdens de presentatie besteden studenten niet alleen aandacht aan hun ontwerp, zij lichten tevens hun analyse toe alsmede hoe het blended onderwijs, inclusief onderwijs waarin generatieve AI is geïntegreerd, te ontwikkelen, te implementeren en te evalueren. Zowel presentatie als ontwerp worden beoordeeld, resulterend in één eindcijfer.

Method of Assessment

De cijfers van studenten voor deze cursus worden berekend op basis van het blended learning ontwerp en het inclusief ontwerp met daarin generatieve AI geïntegreerd en hun presentatie hiervan.

Entry Requirements

Didactiek 1, Praktijk 1 (of vergelijkbaar)

Literature

Wordt bekend gemaakt via Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Custom Course Registration

Alleen studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Innovation, Behavior, Emergence and Markets

Course Code	AM_1052
Credits	3
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E. Muniz Pereira Urias
Examiner	dr. E. Muniz Pereira Urias
Teaching Staff	dr. A.M. van der Geest
Teaching method(s)	Lecture

Course Objective

Innovation, Behavior, Emergence and Markets (IBEM) gives you the theory, tools and practical experience to understand and analyse health care as a complex, adaptive system. You will understand better how individuals and organizations react to changes and interventions and use that to design smarter interventions.

You learn and apply recent system theories to find out why the health care system works as it does: the effects of privatization, survival mechanisms and the "abnormalities" in the health care system. You learn and experience in simulations how individual behavior leads to emergent system effects - and why hidden barriers and unfair earning models obstruct innovation.

You practice and integrate what you have learned in a field work exercise and by studying health care cases. And you experience during the lectures, in the form of small games/simulations, why you (too!) make irrational decisions.

Learning objectives:

- Understand and analyse the general properties of complex adaptive systems.
- Get acquainted with innovation theories in a complex system context.
- Understand and analyze the interactions between individual behavior, economy and the (health care) system.
- Acquire knowledge on innovation, economic and business models against the challenge to improve the health care system.
- Gain insight in what determines the efficiency and effectiveness of health care systems from the viewpoint of a market.
- Improve communication skills in group exercises and presentations.
- Practice system analysis and modelling of health care issues in relation to innovation.
- Get a feeling how negotiations and decision making work in a complex environment.

"Viewing the health care system as a complex system makes a lot of sense" (student evaluation 2020)
"I liked the Field Work Assignment and the Negotiation Gaming, a nice balance between theoretical work and fun simulations/assignments " (student evaluation 2020)
"The practice exam was very nice ! The best practice exam this master. " (student evaluation 2020)

Course Content

IBEM's lectures, exercises, field work and simulations cover topics such as:

- The properties and general laws of complex systems and the mechanisms that determine survival, innovation and cooperation between the actors in the system.
- The relation between 'irrational' behavior and emergent system effects: why customers, hospitals, patients and other actors react in different and unexpected ways, against the assumptions of classical economy.
- Innovation as an on-going, chaotic and competitive process that causes unexpected emergent effects in markets: how should companies, consultants and policy makers deal with this?
- Why network models of systems give insight in social structures and predict how innovations (and diseases!) spread. And explain why it is important to shake hands with Important People.
- Innovations need a 'changeable' environment: learn how to recognize resistance, work on projects at " the edge of chaos" and prevent perverse incentives.
- Companies that understand how the system works use smart strategies for their innovations. New business models, using the experiences of patients and social network models can be effective to support and implement innovations.
- Delivering affordable health to the global population is a real challenge. The World Health Organisation applies a complex system approach to improving health care systems in developing countries.

IBEM combines various scientific disciplines: theory of Complex Adaptive Systems, psychology and behavioral economy, game theory, some innovation theories and (social) network theory.

Additional Information Teaching Methods

IBEM is a periode 3 course (50% work load in four weeks; 3 ECTS).
Total study is 80 hours.

Forms of tuition: lectures, self-study (animation, cases and literature), group exercises, gaming/simulation, presentations/discussions.

Study general CAS theory: 12 hours
Lectures: 12 hours
Field work: 4 hours
Gaming / simulation: 6 hours (of which contact hours/student: 2)
Group exercise: 16 hours
Study articles and cases: 4 hours
Presentation/ innovation market: 4 hours
Study for exam: 20 hours
Exam: 2 hours

Method of Assessment

- Exam (T)(70%)
- Field work group exercise (Prac/V) (pass/fail)
- Case study group exercise (O) (30% of which report (V) 20% and thesis presentation (Pres) 10%).

All three parts should be sufficient/passed - no compensation possible. In case of not passing the exam, one resit opportunity is offered. For an insufficient field work exercise a substitute assignment can be given.

Literature

No text book needed.

All articles and links to supportive educational materials will be
available at Canvas before the course starts.

Additional Information Target Audience

You're a master student (MPA, Innovation Sciences, Something with Business and/or Management, Health Care, Health Economy, Artificial Intelligence, ...)
You have an interest in the interactions between innovation, markets and behavior.
You're eager to understand how "things" work at the system level of health care.
You like to have different, even headstrong perspectives to look at the health care system.

Explanation Canvas

You're supposed to view introductory videos and animations preparing you for the lectures.

Each course module has a discussion page for your questions and a round-off quiz

A responsive mock exam is offered on Canvas which gives you individualized feed back on the status of your learning efforts and shows possible gaps in your preparation.

Recommended background knowledge

No specific requirements

Integrated Biomedical Sciences

Course Code	AM_1281
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.R. Essink
Examiner	dr. D.R. Essink
Teaching Staff	dr. S. Duinkerken, dr. D.R. Essink, V. Avramov MSc, dr. J.M.M. den Haan
Teaching method(s)	Computer lab, Seminar, Study Group, Symposium, Written partial exam, Lecture

Course Objective

- To understand the interdisciplinary nature of biomedical sciences
- To apply interdisciplinary biomedical knowledge to address health issues in society
- To understand the essentials of the immunology and neurobiology of infectious diseases
- To understand infectious disease progression and diagnostics
- To select and interpret research methods and techniques commonly applied in the fields of immunology, neurobiology, infectious diseases, and science and society studies
- To understand and interpret epidemiology of infectious disease outbreaks
- The essentials of statistics used in biomedical sciences
- To understand data collection tools that support outbreak investigations
- To understand outbreak policies and intervention options as well as their limitations
- To develop evidence-informed advice to address an infectious disease outbreak in society
- To communicate and present evidence-informed advice to experts, peers, and society
- To work together with fellow students in groups

Course Content

This is the first course of the Master program Biomedical Sciences. Biomedical sciences range from ‘wet’ laboratory research to the ‘dry’ societal disciplines. In line with the broad biomedical education offered by the MSc Biomedical Sciences, this course aims to show students that the biomedical specializations are not isolated disciplines but are instead integrated with each other, transcending disciplinary boundaries. The central theme of this course is therefore the integration of the different biomedical sciences around infectious diseases, using example diseases/biomedical issues. These diseases form a recurring theme throughout the course, in which they are discussed on different levels: from molecule to society. These are highlighted from the research disciplines immunology, neurobiology, and infectious diseases, as well as from the societal disciplines international public health, science in society, and science communication. In workgroups, students will learn why and how to select research techniques to investigate actual biomedical topics. The interdisciplinary nature of biomedical sciences is further reflected in the group assignment, in which students will work in small groups on applying biomedical knowledge to address relevant health issues in society. Students will use information from the different disciplines to develop evidence-informed advice on an infectious disease outbreak, underlining the interdisciplinary nature of biomedical sciences.

Additional Information Teaching Methods

This course consists of lectures accompanied by workgroups or computer practicals. These will cover two central infectious diseases in all their biomedical aspects, ranging from laboratory research to societal implications.

Additional workgroup hours are scheduled for students to start working on a group assignment. In this assignment, students will work together in groups of approximately 4 students on an infectious disease outbreak in society. Using biomedical peer-reviewed literature, they will investigate the current state of knowledge as well as research questions still to be answered. Students will report on topics relevant to their infectious disease, such as prevalence and incidence, infectivity and spread, tools used in research and diagnostics, and the potential for vaccines and/or therapeutics. Subsequently, students will examine what is needed to develop suitable interventions and outbreak policies to contain further spread of the infectious disease concerned. The end products of this group assignment

will be a report and presentation containing an evidence-informed advice on an infectious disease outbreak in society.

Additional self-study is required for preparation of lectures and workgroups, for the remaining part of the group assignment, and for the final exam.

Method of Assessment

The course content of lectures and work groups will be assessed through an individual exam. The group assignment will be concluded with a report and final presentation. Both the exam and the group assignment each require a 5.5 to pass this course.

Entry Requirements

Entry requirements are similar to the entry requirements of the MSc Biomedical Sciences program. Knowledge of basic immunology (at bachelor level) is compulsory before the start of this course. Students should have conducted a laboratory internship during their bachelor program.

Literature

Lectures, reviews, and scientific research articles will form the primary study material.
Materials will be made available on CANVAS

Additional Information Target Audience

First year master biomedical sciences students

On request and based on motivation and assessment of needed prior knowledge, this course can be chosen as an optional for other master programs

Explanation Canvas

Canvas will be used to share course documents and materials

International Comparative Analyses of Health Care Systems

Course Code	AM_470820
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. J.E.W. Broerse
Examiner	prof. dr. J.E.W. Broerse
Teaching Staff	K.E. Mujica Gastelbondo MSc, prof. dr. J.E.W. Broerse, dr. D.R. Essink
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

At the end of this course students are able to:

- Describe health systems, including its goals and functions, and illustrate how such systems can be analysed, compared, and reformed;
- Recognise the value of using a systems perspective and the complexity of comparing culturally diverse health systems;
- Design methodologies for comparative analyses, including the selection of countries, primary outcomes, conceptual frameworks, and indicators;
- Critically analyse and compare the health systems of two high-income countries (HICs) and two low-and middle-income countries (LMICs) on selected topics, using relevant research questions, publicly available sources, frameworks, outcomes, and qualitative and quantitative indicators;
- Differentiate between comparative health system analyses in HICs and LMICs and identify the unique challenges involved in their comparisons;
- Synthesise findings from comparative analyses in a clear and concise manner (abstracts and indicator

tables);
- Perform an outlier analysis using statistical software (for MSc Global health students only);
- Develop and conduct presentations (lecture and poster presentation) about synthesised comparative results;
- Work as effective team members and collaborate with fellow students in two group assignments;
- Assess the work of your own group and those of other students and give constructive feedback.

Course Content

Until recently, the field of global health mainly embraced disease-specific initiatives and focused on the delivery of health services for specific diseases, as well as their lifestyle, environmental and socioeconomic determinants. After decades of such disease-specific initiatives, it is now recognised that programmes solely targeting the prevention or treatment of specific diseases cannot yield the desired outcomes in the context of weak health systems. Recent pandemics have emphasised the role of health systems in global situations even more. A critical determinant of health, as well as the success of global health initiatives, is the health system as a whole. Recognition of this brought ‘strengthening of health systems’ onto the global health agenda and into funding schemes.

The global health researchers and professionals of this era need to have a good understanding of health systems and the key skills to analyse and compare them in order to; adapt themselves to the different health systems in which they will work; analyse different health systems and identify how they have evolved to this point and in which areas; and determine how they can be strengthened to improve its outcomes.

Serving these purposes, this course aims to equip students with the knowledge and skills related to concepts, theory and methods on the following key areas:

- What a health system is (and different ways of describing it)
- What goals and functions health systems have
- Why and how health systems and their performance can be analysed and compared
- Why and how health systems evolve and can be reformed

In line with these, the course will cover low, middle income countries (LMICs) and high-income countries (HICs) and span across different health system goals (health gain, equity, responsiveness, and effectiveness and efficiency) and functions (stewardship, financing, resource generation, and health services).

In terms of methodology, students learn to use the case study approach for carrying out a comparative analysis. Students will be acquainted with the theory of case studies via lectures and reflection sessions and will implement this method during two group assignments. Both qualitative and quantitative aspects of ‘health systems comparison’ are discussed and critiqued. Furthermore, the complexity and culturally determined nature of health system design and health system reform are presented through a series of lectures form VU-lecturers and experts from a variety of institutions such as the Dutch Royal Tropical Institute (KIT). Case examples and what we can learn from this about (strengthening) health systems will also be discussed.

Small group work for the assignments provides opportunities to practice relevant skills by analysing, first, the health system of two selected HICs according to a defined theme (e.g. mental health, or primary care). Groups will present their findings with a lecture. Subsequently, health systems of two LMICs with comparable financial resources (e.g. GDP per capita, per capita health expenditure, etc.) but different health outputs and outcomes are analysed to find out why two countries are performing in such different ways. The findings will be presented in a poster format. By repeating the analysis on the same theme but with different countries, the students are challenged to constantly improve their own analysis process.

Additional Information Teaching Methods

Teaching methods include lectures by VU teachers and guest lecturers, lectures by other students, workgroups for assignments, problem-driven learning, self-study. The total study time of 168 hours is distributed as follows:

- Lectures ~ 21hrs
- Lectures and practicals on epidemiology & statistics ~19hrs (for MSc Global Health students only)
- Workgroup meetings with coaches ~ 11hrs
- Workgroup work on assignments ~ 58hrs
- Self-study ~ 50 hrs
- Student lectures & poster presentation ~8hrs
- Exam 2,25hrs

Attendance to the workgroup meetings (including feedback sessions), students' lecture presentations and poster presentations are compulsory.

Method of Assessment

The grading of the course is based on Assignment 1, Assignment 2 and the exam:

- Assignment 1: group lecture (pass/fail)
- Assignment 2: group poster presentation (40%)
- Exam: individual assessment (60%)

Each part will be assessed over 100pts and take place on Campus. All parts need to be passed (minimum grade 5.5).

Literature

A selection of literature will be made on the basis of lectures and state of the art research. A selection of last year's literature is listed below.

- Mills & Ranson (2012) The design of health systems.
- World Health Organisation (2011) World Health Report 2000: Health Systems: Improving Performance. (Chapter 1, Chapter 2 and the Statistical Annex)
- Yin RK. Case Study Research: Design and Methods. Sage publications 2018, London. (Chapters 1 and 2)
- World Health Organisation (2006) Working together for health
- Brito Fernandes et al. (2021) Engaging citizens in the development of a health system performance assessment framework: a case study in Ireland Health Research Policy and Systems 19:148
- Plsek PE, Greenhalgh T. Complexity science: The challenge of complexity in health care. Brit Med J. 2001;323(7313):625-8.
- Grin J & Broerse JEW (2017) Introduction. In: Broerse, J.E.W. & Grin J. (eds.) Toward Sustainable Transitions in Healthcare Systems. New York: Routledge
- Van Raak R & de Haan FJ (2017) Key Features of Modern Health Systems: Nature and Historical Evolution. In: Broerse, J.E.W. & Grin J. (eds.) Toward Sustainable Transitions in Healthcare Systems. New York: Routledge
- Fadele KP, et al (2024) Mental health challenges in Nigeria: Bridging the gap between demand and resources. Global Mental Health, 11, e29, 1–5

Additional Information Target Audience

- Compulsory course within the Master specialisation International Public Health
- Compulsory course for 1st year Research Master Global Health students
- Optional course within the Master specialisation Infectious Diseases (master programme Biomedical Sciences).

In any other circumstances admission should be requested from the coursecoordinator.

Recommended background knowledge

It is strongly recommended that students have knowledge on public health and/or public policy in the context of health and/or care.

Internship Biomedical Sciences

Course Code	AM_471158
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.J. van Belle
Examiner	dr. R.J. van Belle
Teaching Staff	K. Wessels MSc, dr. R.J. van Belle
Teaching method(s)	Study Group

Course Objective

The aims of the master's internship are:

- The student learns to independently conduct scientific research.
- The student is able to independently find scientific information and to evaluate this for the benefit of his or her own research question.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.

- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in a scientific report.
- The student is able to cooperate with researchers of various disciplines.
- The student is able to write a scientific report of the research on the level of peer-reviewed academic journals.
- The student is able to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

Course Content

The internship is a compulsory part of the Masters' programme in Biomedical Sciences and involves many different aspects, such as theoretical preparation (literature survey), practical execution (collect data), data-analysis and participation in the scientific activities of a research department. Finally, you present your findings both orally and in a report. The internship has to be preceded by a research proposal.

The internship should have a biomedical research question. Purely observational or inventory projects are not accepted.

For more (detailed) information, please see the placement manual M Biomedical Sciences on Canvas.

Additional Information Teaching Methods

Research project, under supervision of VU-staff.

Method of Assessment

Formative: Within six weeks after the start of the internship a Go/No Go evaluation is made by the VU supervisor. The aim of this interim evaluation is to decide whether the project and the student both have enough potential to continue (Go) or not (No Go). This evaluation is based on:

- Written material by the student: either the Introduction or Methods section of the report or both.
- Attitude of the student and execution of the project during the initial stage.

Summative: The final assessment of the internship is undertaken by the VU-supervisor and the second assessor.

In the final assessment, the VU supervisor assesses four different aspects of the internship:

- the attitude of the student
- the execution of the research (40%)
- the final report (40%): average of both assessors
- the oral presentation (20%)

The second assessor provides an assessment of the final report only.

Only if all marks for each category given by the VU-Supervisor and the Second assessor are a 5.5 or higher, will the thesis/placement be regarded as sufficient. In addition, the student has to pass the category attitude.

Entry Requirements

The student is enrolled in the Research Focus Program of the Master's programme Biomedical Sciences of which the internship is part and has gained at least 30 EC in courses.

The second internship can only start after passing the first internship.

Additional Information Target Audience

Students from the MSc Biomedical Sciences, specifically within the second year Research Focus Program

Additional Information

The Placement Manual is based upon the 'Thesis and Placement Regulations ' of the Faculty of Science and the 'Teaching and Examination Regulations' (TER).

Duration of the internship is one semester (30 EC). It can be extended to 33 or 36 EC (+2 or +4 weeks) if done in the second year, under the condition that total credits for the master won't exceed 120EC.

Your two internships and the literature thesis cannot focus on the same or a highly similar subject, nor can it involve the same supervision and/or techniques. Additionally, the internship must not overlap in time with any courses or the literature thesis.

Custom Course Registration

Internships can commence only after the draft research proposal has been approved by the master's coordinator at least two weeks in advance (assignment on Canvas), uploaded to OnStage, and subsequently approved there.

The Placement Manual and Thesis and Placement Regulations describe the process of completing the internship from the beginning to the final stage.

Explanation Canvas

On Canvas you can find:

- Placement Manual,
- Thesis and Placement Regulations,
- Assignment for the research proposal,
- examples and more.

Internship Immunology

Course Code	AM_471137
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.J. van Belle
Examiner	dr. R.J. van Belle
Teaching Staff	K. Wessels MSc, dr. R.J. van Belle
Teaching method(s)	Study Group

Course Objective

The aims of the master’s internship are:

- The student learns to independently conduct scientific research.
- The student is able to independently find scientific information and to evaluate this for the benefit of his or her own research question.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in a scientific report.
- The student is able to cooperate with researchers of various disciplines.
- The student is able to write a scientific report of the research on the level of peer-reviewed academic journals.
- The student is able to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

Course Content

The internship is a compulsory part of the Masters’ programme in Biomedical Sciences and involves many different aspects, such as theoretical preparation (literature survey), practical execution (collect data), data-analysis and participation in the scientific activities of a research department. Finally, you present your findings both orally and in a report. The internship has to be preceded by a research proposal.

The internship should have an immunological research question. Purely observational or inventory projects are not accepted.

For more (detailed) information, please see the placement manual M Biomedical Sciences on Canvas.

Additional Information Teaching Methods

Research project, under supervision of VU-staff.

Method of Assessment

Formative: Within six weeks after the start of the internship a Go/No Go evaluation is made by the VU supervisor. The aim of this interim evaluation is to decide whether the project and the student both have enough potential to continue (Go) or not (No Go). This evaluation is based on:

- Written material by the student: either the Introduction or Methods section of the report or both.
- Attitude of the student and execution of the project during the initial stage.

Summative: The final assessment of the internship is undertaken by the VU-supervisor and the second assessor.

In the final assessment, the VU supervisor assesses four different aspects of the internship:

- the attitude of the student
- the execution of the research (40%)
- the final report (40%): average of both assessors
- the oral presentation (20%)

The second assessor provides an assessment of the final report only.

Only if all marks for each category given by the VU-Supervisor and the Second assessor are a 5.5 or higher, will the thesis/placement be regarded as sufficient. In addition, the student has to pass the category attitude.

Entry Requirements

The student is enrolled in the Master's programme Biomedical Sciences of which the internship is part.

The student attended the courses:

- AM_1281 Integrated Biomedical Sciences (for students started in 2022-2023 or later)
- AM_470656 Advanced Molecular Immunology and Cell Biology
- AM_470655 Clinical Immunology ór AM_1309 Clin Imm and Dis Mech
- AM_470657 Molecular Infection Biology ór AM_1282 Immunotherapy and Immune Monitoring (or former courses AM_1257 or AM_1031) ór AM_1283 Immunometabolism ór AM_1318 Immune Mod Mon Met

And have acquired 12EC of these courses

The second internship can only start after passing the first internship.

Additional Information Target Audience

MSc Biomedical Sciences students, specialization Immunology

Additional Information

The Placement Manual is based upon the 'Thesis and Placement Regulations ' of the Faculty of Science and the 'Teaching and Examination Regulations' (TER).

Duration of the internship is one semester (30 EC). It can be extended to 33 or 36 EC (+2 or +4 weeks) if done in the second year, under the condition that total credits for the master won't exceed 120EC.

Your two internships and the literature thesis cannot focus on the same or a highly similar subject, nor can it involve the same supervision and/or techniques. Additionally, the internship must not overlap in time with any courses or the literature thesis.

Custom Course Registration

Internships can commence only after the draft research proposal has been approved by the master's coordinator at least two weeks in advance (assignment on Canvas), uploaded to OnStage, and subsequently approved there.

The Placement Manual and Thesis and Placement Regulations describe the process of completing the internship from the beginning to the final stage.

Explanation Canvas

On Canvas you can find:

- Placement Manual,
- Thesis and Placement Regulations,
- Assignment for the research proposal,
- examples and more.

Internship Infectious Diseases

Course Code	AM_471138
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.J. van Belle
Examiner	dr. R.J. van Belle
Teaching Staff	K. Wessels MSc, dr. R.J. van Belle
Teaching method(s)	Study Group

Course Objective

The aims of the master's internship are:

- The student learns to independently conduct scientific research.
- The student is able to independently find scientific information and to evaluate this for the benefit of his or her own research question.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in a scientific report.
- The student is able to cooperate with researchers of various disciplines.
- The student is able to write a scientific report of the research on the level of peer-reviewed academic journals.
- The student is able to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

Course Content

The internship is a compulsory part of the Masters' programme in Biomedical Sciences and involves many different aspects, such as theoretical preparation (literature survey), practical execution (collect data), data-analysis and participation in the scientific activities of a research department. Finally, you present your findings both orally and in a report. The internship has to be preceded by a research proposal.

The internship should have an Infectious Diseases/microbiological research question. Purely observational or inventory projects are not accepted.

For more (detailed) information, please see the placement manual M Biomedical Sciences on Canvas.

Additional Information Teaching Methods

Research project, under supervision of VU-staff.

Method of Assessment

Formative: Within six weeks after the start of the internship a Go/No Go evaluation is made by the VU supervisor. The aim of this interim evaluation is to decide whether the project and the student both have enough potential to continue (Go) or not (No Go). This evaluation is based on:

- Written material by the student: either the Introduction or Methods section of the report or both.
- Attitude of the student and execution of the project during the initial stage.

Summative: The final assessment of the internship is undertaken by the VU-supervisor and the second assessor.

In the final assessment, the VU supervisor assesses four different aspects of the internship:

- the attitude of the student
- the execution of the research (40%)
- the final report (40%): average of both assessors
- the oral presentation (20%)

The second assessor provides an assessment of the final report only.

Only if all marks for each category given by the VU-Supervisor and the Second assessor are a 5.5 or higher, will the thesis/placement be regarded as sufficient. In addition, the student has to pass the category attitude.

Entry Requirements

The student is enrolled in the Master's programme Biomedical Sciences of which the internship is part.

The student attended the courses:

- AM_1281 Integrated Biomedical Sciences (for students started in 2022-2023 or later)
- AM_470656 Advanced Molecular Immunology and Cell Biology (started 2022-2023 or later)
- AM_470657 Molecular Infection Biology
- AM_470127 Containment Strategies of Infectious Diseases ór AM_470052 Parasitology ór AM_1310 The Human Microbiome in H&D

And have acquired 12EC of these courses.

The second internship can only start after passing the first internship.

Additional Information Target Audience

MSc Biomedical Sciences students, specialization Infectious Diseases

Additional Information

The Placement Manual is based upon the 'Thesis and Placement Regulations ' of the Faculty of Science and the 'Teaching and Examination Regulations' (TER).

Duration of the internship is one semester (30 EC). It can be extended to 33 or 36 EC (+2 or +4 weeks) if done in the second year, under the condition that total credits for the master won't exceed 120EC.

Your two internships and the literature thesis cannot focus on the same or a highly similar subject, nor can it involve the same supervision and/or techniques. Additionally, the internship must not overlap in time with any courses or the literature thesis.

Custom Course Registration

Internships can commence only after the draft research proposal has been approved by the master's coordinator at least two weeks in advance (assignment on Canvas), uploaded to OnStage, and subsequently approved there.

The Placement Manual and Thesis and Placement Regulations describe the process of completing the internship from the beginning to the final stage.

Explanation Canvas

On Canvas you can find:

- Placement Manual,
- Thesis and Placement Regulations,
- Assignment for the research proposal,
- examples and more.

Internship International Public Health

Course Code	AM_471139
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.R. Essink
Examiner	dr. R.J. van Belle
Teaching Staff	dr. D.R. Essink
Teaching method(s)	Study Group

Course Objective

The aims of the master's internship are:

- The student learns to independently conduct scientific research.
- The student is able to independently find scientific information and to evaluate this for the benefit of his or her own research question.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in a scientific report.
- The student is able to cooperate with researchers of various disciplines.
- The student is able to write a scientific report of the research on the level of peer-reviewed academic journals.
- The student is able to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

Course Content

The internship is a compulsory part of the one year specialisation as part of the regular master Biomedical Sciences and therefore needs to have a biomedical context.

The internship has to be preceded by a draft research proposal. During the internship, you write an extended research design, collect your data and you do the final analysis. Finally you present your findings both orally and in a report.

Additional Information Teaching Methods

Internship, under supervision of VU-staff.

Method of Assessment

Formative: Within six weeks after the start of the internship a Go/No Go evaluation is made by the VU supervisor. The aim of this interim evaluation is to decide whether the project and the student both have enough potential to continue (Go) or not (No Go). This evaluation is based on:

- Written material by the student: either the Introduction or Methods section of the report or both.
- Attitude of the student and execution of the project during the initial stage.

Summative: The final assessment of the internship is undertaken by the VU-supervisor and the second assessor.

In the final assessment, the VU supervisor assesses four different aspects of the internship:

- the attitude of the student
- the execution of the research (40%)
- the final report (40%): average of both assessors
- the oral presentation (20%)

The second assessor provides an assessment of the final report only.

Only if all marks for each category given by the VU-Supervisor and the Second assessor are a 5.5 or higher, will the thesis/placement be regarded as sufficient. In addition, the student has to pass the category attitude.

Entry Requirements

The student is enrolled in the Master's programme Biomedical Sciences of which the internship is part.

The student attended and passed the courses:

- AM_1255 Research Methods for Health Sciences (ór AM_470817, 6EC, Research Methods for Need Assessments)
- AM_470127 Containment Strategies of Infectious Diseases in Global Context
- AM_470819 Policy, Management and Organization in International Public Health

The second internship can only start after passing the first internship.

Additional Information Target Audience

MSc Biomedical Sciences, specialization International Public Health

Additional Information

The Placement Manual is based upon the 'Thesis and Placement Regulations ' of the Faculty of Science and the 'Teaching and Examination Regulations' (TER).

Duration of the internship is one semester (30 EC).

The internship cannot overlap in time with courses or the literature thesis.

Custom Course Registration

Internships can commence only after the draft research proposal has been approved by the specialization coördinator, followed by approval of the master's coordinator at least two weeks in advance (assignment on Canvas). Subsequently, the proposal is uploaded to OnStage, and will be approved there.

The Placement Manual and Thesis and Placement Regulations describe the process of completing the internship from the beginning to the final stage.

Explanation Canvas

On Canvas you can find:

- Placement Manual,
- Thesis and Placement Regulations,
- Assignment for the research proposal,
- examples and more.

Internship Neurobiology

Course Code	AM_1178
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.J. van Belle
Examiner	dr. R.J. van Belle
Teaching Staff	K. Wessels MSc, dr. R.J. van Belle
Teaching method(s)	Study Group

Course Objective

The aims of the master’s internship are:

- The student learns to independently conduct scientific research.
- The student is able to independently find scientific information and to evaluate this for the benefit of their own research question.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in a scientific report.
- The student is able to cooperate with researchers of various disciplines.
- The student is able to write a scientific report of the research on the level of peer-reviewed academic journals.
- The student is able to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

Course Content

The internship is a compulsory part of the Masters’ programme in Biomedical Sciences and involves many different aspects, such as theoretical preparation (literature survey), practical execution (collect data), data-analysis and participation in the scientific activities of a research department. Finally, you present your findings both orally and in a report. The internship has to be preceded by a research proposal.

The internship should have an neurobiological research question. Purely observational or inventory projects are not

accepted.

For more (detailed) information, please see the placement manual M Biomedical Sciences on Canvas.

Additional Information Teaching Methods

Research project, under supervision of VU-staff.

Method of Assessment

Formative: Within six weeks after the start of the internship a Go/No Go evaluation is made by the VU supervisor. The aim of this interim evaluation is to decide whether the project and the student both have enough potential to continue (Go) or not (No Go). This evaluation is based on:

- Written material by the student: either the Introduction or Methods section of the report or both.
- Attitude of the student and execution of the project during the initial stage.

Summative: The final assessment of the internship is undertaken by the VU-supervisor and the second assessor.

In the final assessment, the VU supervisor assesses four different aspects of the internship:

- the attitude of the student (pass/fail)
- the execution of the research (40%)
- the final report (40%): average of both assessors
- the oral presentation (20%)

The second assessor provides an assessment of the final report only.

Only if all marks for each category given by the VU-Supervisor and the Second assessor are a 5.5 or higher, will the thesis/placement be regarded as sufficient. In addition, the student has to pass the category attitude.

Entry Requirements

The student is enrolled in the Master's programme Biomedical Sciences of which the internship is part.

The student attended the courses:

- AM_1281 Integrated Biomedical Sciences (for students started in 2022-2023 or later)
- AM_1275 From Molecule to Mind
- AM_1005 Clinical Neurosciences
- AM_1214 Genetics in Neuroscience

And have acquired 12EC of these courses.

The second internship can only start after passing the first internship.

Additional Information Target Audience

MSc Biomedical Sciences students, specialization Neurobiology

Additional Information

The Placement Manual is based upon the 'Thesis and Placement Regulations ' of the Faculty of Science and the 'Teaching and Examination Regulations' (TER).

Duration of the internship is one semester (30 EC). It can be extended to 33 or 36 EC (+2 or +4 weeks) if done in the second year, under the condition that total credits for the master won't exceed 120EC.

Your two internships and the literature thesis cannot focus on the same or a highly similar subject, nor can it involve the same supervision and/or techniques. Additionally, the internship must not overlap in time with any courses or the literature thesis.

Custom Course Registration

Internships can commence only after the draft research proposal has been approved by the master's coordinator at least two weeks in advance (assignment on Canvas), uploaded to OnStage, and subsequently approved there.

The Placement Manual and Thesis and Placement Regulations describe the process of completing the internship from the beginning to the final stage.

Explanation Canvas

On Canvas you can find:

- Placement Manual,
- Thesis and Placement Regulations,
- Assignment for the research proposal,
- examples and more.

Internship Science in Society

Course Code	AM_1133
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M.B.M. Zweekhorst
Examiner	dr. R.J. van Belle
Teaching Staff	prof. dr. M.B.M. Zweekhorst
Teaching method(s)	Study Group

Course Objective

The aim of the internship as part of the**specialization Science in Society within the****Master Biomedical Sciences of VU Amsterdam** is to apply the competences acquired during

the previous courses in a research project in order to ground the knowledge, attitudes and skills of interdisciplinary research. More specifically, the aims of the internship are:

- The student learns to independently conduct scientific research.
- The student is able to independently find scientific information and to evaluate this for the benefit of their own research question.
- The student is able to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student is able to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in a report.
- The student is able to cooperate with researchers of various disciplines.
- The student is able to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

Course Content

The internship is a compulsory part of the one year specialisation as part of the regular master Biomedical Sciences at VU University Amsterdam and therefore needs to have a **biomedical context**.

An internship placement must provide the student with the opportunity to learn how to conduct research under supervision. The on-site supervisor of the internship is linked to an academic or research institution.

Internships can be done at various locations such as the Ministry of Health, Welfare and Sports, the Public Health Inspectorate, the Health Council, medical organizations such as the municipality health service (GGD), consultancies, the (pharmaceutical) industry and several research institutes, such as universities or e.g. the National Institute for Public Health and the Environment (RIVM).

An internship typically has three phases

- In the first phase, you write your final research proposal consisting of an introduction, background, theoretical/conceptual framework, research questions and your research methodology.
- In the second phase, you collect your (qualitative and/or quantitative) data.
- In the third phase, you do your final analysis and present your findings both orally and in a report. The presentation seminar is a compulsory part of this third phase.

Additional Information Teaching Methods

Internship, under supervision of VU-staff.

Method of Assessment

Report (40%), Oral presentation (20%), Execution (40%) and Attitude (Pass/fail)

Within six weeks after the start of the master internship, an interim evaluation (Go/No Go) will take place to assess whether there is a reasonable chance of the internship being brought to a successful completion.

The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam. The daily supervision can be carried out by a PhD-student, postdoc or researcher.

Entry Requirements

To ensure that students do have enough background knowledge, it is required that you have passed the three compulsory courses:

- AM_1182 Research Methods for Analyzing Problems
- AM_470571 Analysis of Governmental Policy
- AM_470572 Communication, Organization and Management.

The second internship can only start after passing the first internship.

Additional Information Target Audience

Only students MSc Biomedical Sciences **VU**, specialization Science in Society.

Exchange students and students from other MSc programmes/universities, please use the course code AM_1185.

Additional Information

The Placement Manual is based upon the 'Thesis and Placement Regulations ' of the Faculty of Science and the 'Teaching and Examination Regulations' (TER).

Duration of the internship is one semester (30 EC).

The internship cannot overlap in time with courses or the literature thesis.

Custom Course Registration

Internships can commence only after the draft research proposal has been approved by the specialization coördinator, followed by approval of the master's coordinator at least two weeks in advance (assignment on Canvas). Subsequently, the proposal is uploaded to OnStage, and will be approved there.

The Placement Manual and Thesis and Placement Regulations describe the process of completing the internship from the beginning to the final stage.

Explanation Canvas

On Canvas you can find:

- Placement Manual
- Thesis and Placement Regulations
- Assignment for the research proposal
- examples and more

Literature Review MPA

Course Code	AM_1251
Credits	6
Period	P1, P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.H.J. van de Voort
Examiner	dr. S.H.J. van de Voort
Teaching Staff	K.E. Mujica Gastelbondo MSc, dr. S.H.J. van de Voort

Teaching method(s)	Study Group, Lecture

Course Objective

- Identify and describe different approaches to conduct literature reviews and their strengths & weaknesses.
- Formulate a relevant review question and select an appropriate approach to answer it
- Comprehensively search the literature using appropriate keywords and databases
- Critically appraise and analyze literature relevant to the chosen research question.
- Analyze and synthesize the key findings of the studies included in the review so that the review question is answered.
- Critically discuss the outcome of the review in light of its contribution to the existing knowledge base and its societal implications
- Gain skills in writing a literature review report

Course Content

The literature study is an important, compulsory part of the Master's Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences (MPA). In this course, students acquire knowledge, skills and competences which allow them to conduct a literature review in the field of Health and Life Sciences based Management and Entrepreneurship, Health and Life Sciences based Policy, Health and Life Sciences based Communication or International Public Health.

The literature review has to focus on a research question related to the topics covered in the master's program. In groups, students choose a topic of their interest from a list provided by the course coordinator
at the start of the course.

The lectures cover the basic conceptual and operational elements necessary to conduct a (systematic) literature review. Then, those elements are practised and discussed during the working group sessions. Additionally, each group of students is guided by an external lecturer during the literature review design and writing process.

Attendance for the working group sessions is compulsory. Students can only miss one session. In this case, who miss one working group needs to hand in an extra assignment related to the content of that particular session.

Additional Information Teaching Methods

Lecture; Working group (working group attendance is compulsory)

Method of Assessment

Individual (30%), Groupwork (70%)
All grade elements must be sufficient (minimal 5,50) to pass the course.

Literature

It will be announced on Canvas

Literature Study M Biomedical Sciences

Course Code	AM_1319
Credits	6
Period	P1, P3
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.J. van Belle
Examiner	dr. R.J. van Belle
Teaching Staff	K. Wessels MSc, dr. R.J. van Belle
Teaching method(s)	Seminar, Study Group, Lecture

Course Objective

The student is able to:

- formulate a research question which can be answered by studying literature

- comprehensively search the literature using appropriate keywords and databases
- critically appraise and analyze literature relevant to the chosen research question
- synthesize and interpret the key findings of the literature so that the research question is answered
- place the findings in a broader context
- write a scientific report of the results in scientific English at the level of peer-reviewed academic journals.
- critically evaluate their contribution, reflect on their learning, and identify areas for improvement.

Course Content

The literature study is an important, compulsory part of the Master's programme in Biomedical Sciences. In this course, students acquire knowledge, skills and competences which allow them to conduct a literature study in their biomedical field.

In groups, students choose a topic of their interest from a list provided at the start of the course.

The lecture covers the aim and setup of the course and aims to refresh your background knowledge. The workshops cover the basic elements to conduct a (systematic) literature study and allows the group to work on the report. Alongside, each group of students is guided by a topic supervisor.

Attendance for the lecture and most workshops is compulsory.

Additional Information Teaching Methods

Group project. Support through workshops by the lecturers and content discussions with your topic supervisor.

Method of Assessment

The final report (90%)
Reflection assignment (10%)

Both grades have to be 5.5 or higher.

Literature

Will be published on Canvas

Additional Information Target Audience

Only for second year MSc Biomedical Sciences students

Custom Course Registration

This courses takes place in either period 1 (Sep-Oct) ór in period 3 (January).

Explanation Canvas

Groups/topics in Canvas.

Recommended background knowledge

We recommend that you have at least completed your first year courses and internship.

Literature thesis Biomedical Sciences

Course Code	AM_471135
Credits	9
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.J. van Belle
Examiner	dr. R.J. van Belle
Teaching Staff	dr. R.J. van Belle
Teaching method(s)	

Course Objective

The literature thesis has the aim to reflect on scientific literature, based on a biomedical research question within your biomedical disciplinairy specialization.

In a total of 6 weeks (full time), the student specializes in a certain biomedical topic by gathering and analyzing (recent) scientific articles and other literature that can be included in the literature research.

Course Content

A literature thesis is a compulsory part of the Master's programme in Biomedical Sciences.

The student can decide on the topic of the thesis, as long as it is focussed on one of the biomedical disciplinary research specialisations they have completed (Immunology, Infectious Diseases or Neurobiology) and should have a biomedical research question.

Additional Information Teaching Methods

Literature thesis, under supervision of VU/AmsterdamUMC-staff.

Method of Assessment

The assessment of the literature thesis is undertaken by the
VU-supervisor.
The literature thesis is assessed on the following aspects:

- execution of the thesis
- final report
- oral presentation

Entry Requirements

We recommend that you have at least completed all specialization courses, and preferably also the internship of the research specialization of which this thesis is part of.

Additional Information Target Audience

A literature assignment is compulsory for the MSc Biomedical Sciences students

Additional Information

Detailed information can be found in the Guidelines (Canvas), based upon the 'Thesis and Placement Regulations ' of the Faculty of Science and the 'Teaching and Examination Regulations' (TER).

Your literature thesis and your two internships cannot focus on the same or a highly similar subject, nor can it involve the same supervision. Additionally, the literature thesis cannot overlap in time with a fulltime load of courses or the internship.

Custom Course Registration

The literature thesis can can commence only after the draft research proposal has been approved by the master's coordinator at least two weeks in advance (assignment on Canvas), uploaded to OnStage, and subsequently approved there.

Explanation Canvas

On Canvas you can find:

- Guidelines for the literature thesis,
- Assignment for the research proposal,
- examples and more.

Management of Innovative Technologies in Community-Based Health Care

Course Code	AM_1181

Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R. Baumgartner
Examiner	dr. R. Baumgartner
Teaching Staff	
Teaching method(s)	Study Group, Lecture

Course Objective

The learning goals of MITCH are:

- To learn about the development and usage of M-health, E-health, point of care diagnostics, self-monitoring tools, (big) data analysis tools and artificial intelligence.
- To give in depth background of the technological developments, their usage and impact on health care professions.
- To familiarize with current and upcoming technologies in community based health care.
- To reflect on the (possible) consequences of technological developments for patients, care providers and the health care system at large.
- To be able to put these developments in context and reflect critically on practical issues from specific theoretical perspectives.
- To engage with care providers, patient groups, technology providers and others to learn about and discuss "hot topics and developments"

Course Content

The MITCH course provides students with an in-depth study of (the development of) front-line innovative technologies in community based health care. The current changes in health care systems and health care provision include a central role for technology. With increasing demands on staff and resources, technology may contribute to sustainable solutions in health care. In this course we expose students to front-line technologies and challenge them to reflect on the consequences that new technologies in health care have.

Students will not only focus on a technical understanding of the developments, but also on how technology can change the way health care is delivered to improve the healthcare system as a whole. This topic will be approached based on case studies from different perspectives, including patients, health consumers, health care providers/entrepreneurs, insurers/financiers and scientists / technology developers.

Additional Information Teaching Methods

MITCH is a course of eight weeks (6 ECTS). The total study time is 160 hours. Tuition methods include interactive lectures, workshops, and self-study. In the course we will make use of face to face teaching and blended learning when needed.
The different elements have the following study time:

- Lectures (h): 24 hours (24 contact hours/student)
- Work groups (w): 20 hours (20 contact hours/student)
- Group assignment (pro): 60 hours
- Preparing the presentation and the presentation seminar (pro): 4 hours (2 contact hours/student)
- Self-study: 50 hours
- Exam: 2 hours

Method of Assessment

The final grade for MITCH is composed of:
- Written exam (T) (40 %)
- Group assignment (V) (40 %)
- Presentation of the assignment in a seminar form (Pres) (20 %)

All parts need to be passed, no compensation for an insufficient result for the exam or assignment report. For an insufficient presentation, a replacing assignment can be given. For the exam there is a resit opportunity.

Literature

To be announced and provided on Canvas.

Additional Information Target Audience

MITCH is a compulsory course for the MPA specialization CHT and an elective course for the MPA specialization M&E. MITCH is considered a Science Course for all MPA specializations.

Explanation Canvas

Detailed course information and schedules will be available in time on Canvas before the course starts.

Molecular Infection Biology

Course Code	AM_470657
Credits	6
Period	P2
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. Bitter
Examiner	prof. dr. W. Bitter
Teaching Staff	prof. dr. W. Bitter, A. Speer, drs. C.P. Kuijl
Teaching method(s)	Seminar, Practical, Study Group, Written partial exam, Lecture

Course Objective

- To understand how the pathogen interacts with its host and how this is studied: in vitro studies, (alternative) in vivo models, different approaches of mutant screening
- To understand the genetic variation of microbial pathogens (viruses and bacteria) and the effect this variation has on host adaptation.
- To understand what virulence factors are and how they are identified and regulated by the pathogen.
- To apply the acquired knowledge to interpret scientific literature and scientific hypotheses regarding host-pathogen interactions.

Course Content

The rapid expansion of genomic, proteomic, and RNA sequencing data from both microbial and eukaryotic hosts, combined with continuous advances in molecular biology, has enabled detailed analysis of the molecular interactions between pathogens and their hosts. This knowledge is essential, as we are constantly confronted with emerging infectious agents and the re-emergence of historical diseases, necessitating the development of new vaccines and antimicrobial therapies.

A central challenge lies in selecting the most appropriate techniques for addressing specific research questions and interpreting potentially conflicting results arising from different experimental approaches, i.e. the choice of the technique or approach used already determines which factors you can and can not identify .

This course is designed to provide a comprehensive understanding of molecular biology as it pertains to infectious agents. It focuses on the application of molecular biology to investigate host-pathogen interactions of pathogenic bacteria and viruses—with an emphasis on virulence factors as well as the genetic susceptibility of hosts to infection. The course aims to equip students with the specialized knowledge and analytical skills required to critically evaluate primary literature in the field of molecular medical microbiology.

Additional Information Teaching Methods

The course has three different parts: lectures, literature presentations and rebuttal assignment. In the literature assignment students will use their acquired knowledge from the lectures to dig deeper into specific aspects of host-pathogen interactions based on a recent high-profile publication. In the rebuttal assignment the students will work on an actual scientific publication that has come back from the reviewers to interpret the comments and use these to write a rebuttal using new data.

Contact hours:
Lectures: 20h
Literature Workshop: 16-20h (depending on number of students)
Rebuttal assignment: 5 h

Method of Assessment

- written exam (50% of final mark and should be minimally 5.5)
- literature discussion (workshop, 25% of final mark)
- practicals/assignment (25% of final mark)

Entry Requirements

Bachelor's course 'Infectious Diseases' and 'Immunology' or an equivalent course in Microbiology and Molecular Biology.

Literature

List of important articles that are supporting the lectures will be provided on CANVAS

Additional Information Target Audience

Students with a keen interest to study the interaction between a pathogen and its host, from a practical as well as a theoretical point of view. Compulsory course for specialization Infectious Diseases.

Additional Information

Guest lectures:
Dr. Lia van der Hoek, AUMC Amsterdam, identification of novel viral pathogens
Dr. Wiep Klaas Smits, LUMC Leiden, working with gut pathogens

Custom Course Registration

You need to register yourself for the course. The course coordinator will register you for the practicals/assignments.

Parasitology

Course Code	AM_470052
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.R. van Dijk
Examiner	dr. M. van der Hoeven
Teaching Staff	dr. M. van der Hoeven, dr. M. Campos Ponce, dr. M.R. van Dijk, dr. J.R. Haanstra
Teaching method(s)	Study Group, Practical, Lecture

Course Objective

This course aims to provide students with a wide knowledge and understanding of medical parasitology. The course focuses on:

- Parasites and their virulence;
- Transmission;
- The role of parasites and host(s) on the outcome of disease;
- Immunological aspects;
- (Laboratory) diagnosis of parasites;
- Epidemiology, treatment, prevention and control of parasitological diseases.

Course Content

This course will focus on Medical Parasitology.

The lectures will provide an overview of parasites that are of medical importance, focussing on:
parasite life cycles; interactions between parasites and their host(s) (e.g. invasion, survival, immune modulation and underlying molecular mechanisms);
diagnosis (development and application); hygiene hypothesis; epidemiology; treatment; prevention (including vaccines); control and elimination.
The majority of the lectures will be given by some of the top researchers and professors within the field of parasitology in the Netherlands, Belgium and Unganda.
These expert guest speakers will provide you with a broad overview of what is happening in this field.

The lectures will be interactive and should therefore be prepared by reading the corresponding literature. Your input in the lectures will be of utmost importance.

Specific teaching goals:

After this course you will have learned the principles of medical parasitology and will be able to:

- Apply these principles to different parasite groups;
- Describe parasite life cycles using correct terminology;
- Identify the role of the host and parasite (virulence factors) on the outcome of an infection and disease;
- Describe the underlying molecular mechanisms involved in parasite infection;
- Describe different diagnostic techniques and their respective advantages and disadvantages;
- Describe and discuss the principles for treatment, prevention, control or eradication programmes for parasite infections of medical importance;
- Analyse, discuss and present literature in the parasitology research field;
- Make suggestions for future parasitological research

Furthermore, this course includes academic and professional skills development in all five areas of competence: 1) communication skills, 2) ethical skills, 3) teamwork- and interpersonal skills, 4) problem-solving and analytical skills, and 5) technical/computer skills.

Additional Information Teaching Methods

This course will include interactive guest lectures covering all aspects of medical parasitology. This will be followed by discussion groups, peer learning by using the online tool Perusall, and in-class assignments, in which you will be expected to demonstrate an in-depth understanding of the course content and literature. You will also have the opportunity to identify several parasites in a practical session. In the final week you will (within a group) present a grant application on a parasite of choice during an elevator pitch or prepare and actively participate in a debate on a selected parasitological topic. All workgroup-sessions are compulsory.

An written examination will take in week 7. Lectures: 31 hours Workgroups: 20 hours Practical parasite identification: 4 hours

Method of Assessment

You will pass the course if you have attended all compulsory sessions, if all assignments are sufficient and the exam is above 5,5 (so not 5,4!).

Entry Requirements

Solid knowledge on basic cell biology, basic immunology and the basics of infectious diseases is compulsory before the start of the course.

Literature

On Canvas

Additional Information Target Audience

MSc programme Health Sciences, specialization Infectious diseases and Public Health, optional course within the MSc programmes of Biomedical Sciences.

Additional Information

Several guest lecturers will be invited to teach on Medical Parasitology.

This course will be offered for the final time within the MSc Health Sciences in the 2025–2026 academic year.

Explanation Canvas

All communication will be via Canvas. It is recommended to check Canvas regularly for the latest updates.

Pedagogische en Didactische verdieping Semester 1

Course Code	O_MPEDIVER_F
Credits	9

Period	P1+2+3
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	dr. B. de Vries, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, drs. A.J.C. Monquil, dr. I. Pauw, dr. H.B. Westbroek, dr. O.P.M. van Buuren, drs. M.M. Bout, drs. J.K.W. Riksen, drs. E.D. van Noort, drs. Y.G. Meindersma, drs. N.H. Ypenburg, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, dr. S.M. Calor, C.L. Geraedts, M. Beukering, drs. R. Cornelisse MSc, drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, dr. S. Donszelmann
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren, (7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Pedagogische en Didactische Verdieping Semester 2

Course Code	O_MPEDIVER
Credits	9
Period	P4+5+6
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts

Teaching Staff	drs. M.M. Bout, drs. R. Cornelisse MSc, C.L. Geraedts, dr. H.B. Westbroek, M. Beukering, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, dr. B. de Vries, dr. C.D.P. van Oeveren, drs. L.J. van Well-van Grootheest, drs. E.D. van Noort, dr. S.M. Calor, J. Quartel MA, N. el Abdellaoui MSc, prof. dr. M. Meeter, dr. I. Pauw, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. S. Donszelmann, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. J.K.W. Riksen, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren, (7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback..

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Peergroup 1 Semester 1

Course Code	O_MLPEERGR_1
Credits	0
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef en de basisproef, en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor dit onderdeel ingeschreven te staan.

Peergroup 1 Semester 2

Course Code	O_MLPEERG_1F
Credits	0
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	N. el Abdellaoui MSc, drs. Y.G. Meindersma, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef (Didactiek 1) en de basisproef (Didactiek 2), en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 1

Course Code	O_MLPEERG_2F
Credits	0
Period	P1+2+3
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit, N. el Abdellaoui MSc, drs. Y.G. Meindersma
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Aantal werkgroepen op maandag (instituutsdag).

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 2

Course Code	O_MLPEERGR_2
Credits	0
Period	P4+5+6
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. R. Cornelisse MSc, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Werkgroepen.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Policy, Management and Organisation in International Public Health

Course Code	AM_470819
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.O. Kok
Examiner	dr. M.O. Kok
Teaching Staff	dr. M.O. Kok, dr. A.M.C. Loeber
Teaching method(s)	Study Group, Lecture

Course Objective

- To develop a detailed understanding of the health policy process and its outcomes both at national and international level
- To acquire insight into different theoretical concepts on policy design in the field of public health
- To understand how policy decisions in the health sector are translated into concrete programs and projects, and subsequently implemented
- To get acquainted with different management and governance practices in health programs
- To get acquainted with and acquire skills in international health diplomacy, resolution writing, negotiation and the UN procedures of the World Health Organization

Course Content

This course contains two core parts that will run parallel throughout the course: a theoretical part and a practical, diplomacy, part. In the theoretical part you study different theoretical concepts of policy science in international public health. You study core concepts of public administration in relation to IPH such as power relations, securing public interest, public versus private sector, managing change and the network society. Questions are addressed such as: In what way does the political structure of a country influence health policies; Why do certain topics get on the policy agenda while other topics never make it; Why do policy makers and politicians regularly seem to ignore scientific insights; To what extent do international organizations (such as the World Bank and the World Health Organization) influence national policies? To link these the theoretical insights to concrete examples, all students will have to read the book 'Wisdom of Whores', which traces the response to the HIV/AIDS epidemic. In the diplomacy part you develop basic diplomatic skills by practicing them in 4 training sessions and a final 1.5 day World Health Organization simulation under Model United Nations rules of procedure (WHO MUN). Model United Nations (informally abbreviated as Model UN or MUN) is an academic simulation of the United Nations that aims to educate you about civics, effective communication, globalization and multilateral diplomacy. In Model UN, you take on roles as foreign diplomats and participate in a simulated session of the WHO.

Additional Information Teaching Methods

Lectures (32 hours), training workshops (16 hours) and simulation (12 hours), self study (102,5 hours), and examination (2.5 hours)

Method of Assessment

Individual exam (70%) and diplomacy assignment (30%). Both grades need to be at least 5.5 to pass the course.

Literature

- "Making Health Policy", Kent Buse,Nicholas Mays & Gill Walt, 2005, Open University Press, ISBN 0-335-21839-3
- "The Wisdom of Whores", Elizabeth Pisani, 2008, Granta Publications, ISBN978-1-84708-076-9
- Other reading materials via Canvas

Additional Information Target Audience

Compulsory course within the Master specialization International Public Health; optional course for students in other specializations of the Masters Health Sciences and Biomedical Sciences.

Additional Information

Attendance of training workshops and simulation is compulsory.

For further information and application, please contact Maarten Kok (m.o.kok@vu.nl)

This course will be offered for the final time within the MSc Health Sciences in the 2025–2026 academic year.

Policy, Politics and Participation

Course Code	AM_470589
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A.M.C. Loeber
Examiner	dr. A.M.C. Loeber
Teaching Staff	A. Teijeiro Fokkema, V. Avramov MSc, dr. A.M.C. Loeber
Teaching method(s)	Study Group, Lecture

Course Objective

On successful completion of this course, you will be able to:

1. analyse a complex societal problem, taking into consideration stakeholder perspectives;
2. apply models and insights from political theory and the policy sciences to make sense of the problem;

3. apply specific qualitative social science methods for data collection (focus group approach) and reflect on possibilities for working in a transdisciplinary manner;
4. develop an argument, combining conceptual and empirical insights, on how the problem might be resolved;
5. convey your assessment of the problem and its solution in a convincing manner, improving your communication skills;
6. work effectively in a project team, improving skills in collaborative project design and execution.

Course Content

In this course, you will combine theory with practice to make sense of a concrete societal issue and contribute to its resolution. In a small group, you will learn to apply a specific method for data collection (focus group discussions) which will help you include the views of those affected by the issue in your process of articulating the issue as a (policy) problem, and outlining directions for its solution. The course pays attention to skills required in transdisciplinary research.

The course focuses on societal problems and their solutions, of which the complexity can be mind-blowing. Take for instance the relation between someone's health and diet: if a person's eating habits are detrimental to their health, what to take on board when thinking about a solution? Does that mean developing an information campaign about buying healthy food, or rather creating a tax on foods that are high in fat, salt or sugar? Or should one look for solutions by changing the design of supermarkets or even city centers? In the process of problem articulation, on a conceptual level, we will draw on the policy sciences, participation literature and political theory to answer such questions as: whose responsibility is it to act on a problem, is it an individual’s responsibility, that of business, and what is the role of government? We will look particularly in the relation between power and knowledge in developing solutions: whose expert views should come to bear on the policy decisions to be taken, that of academically trained experts, or should we also include the views and expertise of those who are literally impacted by the issue - the stakeholders - and if so, how and to which extent?

On a practical level, you will engage in conversations with people who are active in the field of policy-making and societal change, who will share their experiences as guest lecturers. And you will be trained to apply a research method - focus groups - that will get you in conversation with stakeholders. Specific attention is paid to the research competencies and personal skills necessary for engaging in such form of social science data collection. At the end of the course, you will present your findings and recommendations verbally.

Additional Information Teaching Methods

The course is designed from a 'Student-in-the-Lead' perspective, seeking to build on and stimulate participants' sense of ownership of their own learning trajectory. This means that it is based on intensive exchanges among students, between students and (guest) lecturers, and between students and stakeholders. Therefore, participation in the (guest) lectures, which are integrated together with practical discussion assignments into working seminars, is mandatory, as is participation in the focus group seminars and literature discussion sessions. By thus making the most of our time together, we aspire to combine a reflective, in-depth way of approaching our themes and topics with a time-efficient way of working.

Method of Assessment

Progress assessment and examination will be as follows:

Individual grade [50%]:

- Literature discussion facilitation and participation (20%)
- (open book) Exam (30%)

Group grade [50%]:

- Focus group Design (20%)
- Presentation (including analysis, policy recommendations) (30%)

For the group assignments a pass grade (> 5.5) needs to be obtained in order to receive a final mark. If you fail the exam, a resit will be arranged.

Entry Requirements

Basic knowledge of the policy process (via AGP or an equivalent introductory course on policy) and of basic research skills is required.

Literature

To be announced on Canvas

Additional Information Target Audience

Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences. Mandatory course for MPA students who specialize in Policy.

Additional Information

Attendance is compulsory. (You will spend a great deal of your time on team work.)

Custom Course Registration

Registration deadline by VUnet is 4 weeks before the start of the course.

Practice 1

Course Code	O_MLPRAK_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, N. el Abdellaoui MSc, drs. E.D. van Noort, M. Beukering, dr. H.B. Westbroek, C.L. Geraedts, dr. O.P.M. van Buuren, dr. S.M. Calor, drs. R. Cornelisse MSc, dr. S. Donszelmann, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, drs. Y.G. Meindersma, drs. M.M. Bout, dr. I. Pauw, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, drs. A.J.C. Monquil, dr. B. de Vries
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 1

Course Code	O_MLPRAK_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. C.D.P. van Oeveren, M. Beukering, drs. M.M. Bout, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. H.R. Goudsmit, drs. J.K.W. Riksen, dr. H.B. Westbroek, drs. R. Cornelisse MSc, dr. S. Donszelmann, N. el Abdellaoui MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, dr. I. Pauw, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, dr. O.P.M. van Buuren, C.L. Geraedts, prof. dr. M. Meeter, drs. E.D. van Noort, dr. S.M. Calor, drs. M.A. Huijskes MA
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op

het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. I. Pauw, prof. dr. M. Meeter, dr. H.B. Westbroek, M. Beukering, dr. C.D.P. van Oeveren, C.L. Geraedts, D.S. Hoefakker-Laanstra, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. B. de Vries, dr. S.M. Calor, drs. A.J.C. Monquil, drs. J.K.W. Riksen, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. M.A. Huijskes MA, drs. E.D. van Noort, dr. O.P.M. van Buuren, dr. S. Donszelmann
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de module geldt:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

Voor de master geldt:

De werkplekbegeleider beoordeelt Praktijk 2 aan de hand van het Beoordelingsformulier Praktijk.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Additional Information

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2
Credits	9
Period	P2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil

Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. A.J.C. Monquil, M. Beukering, drs. E.D. van Noort, dr. H.B. Westbroek, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, C.L. Geraedts, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, dr. B. de Vries, dr. I. Pauw, drs. H.R. Goudsmit, dr. S.M. Calor, dr. O.P.M. van Buuren, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, prof. dr. M. Meeter, drs. M.A. Huijskes MA, drs. N.H. Ypenburg
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor een modulestudent geldt het volgende:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

Voor een masterstudent geldt het volgende:

Praktijk 2 wordt beoordeeld door de werkplekbegeleider.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3F
Credits	15
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. H.R. Goudsmit, dr. B. de Vries, D.S. Hoefakker-Laanstra, C.L. Geraedts, drs. R. Cornelisse MSc, drs. Y.G. Meindersma, R.B. van der Vos, drs. N.H. Ypenburg, M. Beukering, drs. A.J.C. Monquil, drs. E.D. van Noort, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, dr. O.P.M. van Buuren, dr. I. Pauw, dr. C.D.P. van Oeveren, R.M.C. Canisius, drs. M.M. Bout, drs. J.K.W. Riksen, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter, dr. H.B. Westbroek, dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. Ook met onderwerpen die bij keuzevakken op het instituut ter sprake komen zal in de praktijk worden geëxperimenteerd. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3
Credits	15
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. E.D. van Noort, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, dr. S.M. Calor, dr. H.B. Westbroek, dr. O.P.M. van Buuren, M. Beukering, N. el Abdellaoui MSc, dr. B. de Vries, drs. M.M. Bout, drs. R. Cornelisse MSc, drs. A.J.C. Monquil, dr. C.D.P. van Oeveren, prof. dr. M. Meeter, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, C.L. Geraedts, drs. J.K.W. Riksen
Teaching method(s)	Lecture

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Praktijk 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Praktijkonderzoek

Course Code	O_MLPROZ
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. H.B. Westbroek
Examiner	dr. H.B. Westbroek
Teaching Staff	dr. H.B. Westbroek
Teaching method(s)	Study Group, Lecture

Course Objective

Aan het eind van deze cursus heeft de student:

1. Kennis van en inzicht in het opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek;
2. Vaardigheid in het in samenwerking met experts op het instituut en op school opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek.

Concreet houdt dit in:

- Het vertalen van een praktijkprobleem naar een onderzoeksvraag;
- Het ontwerpen en uitvoeren van een kleinschalige dataverzameling;

- Het maken van een analyseplan en uitvoeren van (kwalitatieve en/of kwantitatieve) analyse;
- Het rapporteren van de gebruikte methode, de resultaten, conclusie en discussie in een presentatie.

Course Content

In het keuzevak Praktijkonderzoek staat het leren toepassen van onderzoeksvaardigheden in de onderwijspraktijk centraal. In dit keuzevak leer je een actieonderzoek opzetten en uitvoeren ten behoeve van je eigen professionalisering. Vaak wordt een professionaliseringsvraag breder gedeeld in je sectie/school. Actieonderzoek is een vorm van onderzoekend leren. Je gebruikt theorie en bevindingen uit onderwijsonderzoek om jouw praktijkprobleem te analyseren en om ontwerp/voornemens te formuleren. Je leert hoe je in je klassen data kan verzamelen over de invloed van jouw ontwerp, op de leerprocessen van je leerlingen.

Additional Information Teaching Methods

introductiecollege (1,5 uur), werkgroepen/intervisiesessies (7,5 uur), zelfwerkzaamheid en individuele begeleidingsgesprekken (157,5 uur), presentaties (1,5 uur).

Method of Assessment

Praktijkonderzoek wordt afgesloten met een presentatie. Tussentijds dient ook een onderzoeksplan te worden ingeleverd dat wordt beoordeeld (go/no go).

Entry Requirements

Vereiste voorkennis: Didactiek 2, Praktijk 1 en onderzoekservaring op bachelor/masterniveau in het eigen domeinvak.

Literature

- Literatuur uit Didactiek 1 en Didactiek 2 vormt een basis.
- Literatuur wordt aangeboden in de vorm van wetenschappelijke artikelen (CANVAS).
- Van studenten wordt verwacht dat ze aanvullend zelf literatuur zoeken en gebruiken om hun theoretisch kader en onderzoeksmethode te kunnen uitwerken en onderbouwen.

Additional Information Target Audience

Praktijkonderzoek is een van de specialisaties binnen de ULO lerarenopleiding.

Explanation Canvas

Alle informatie over de opzet van het vak, de bijeenkomsten, toetsing en literatuur wordt aangeboden via CANVAS. Communicatie gebeurt via CANVAS (discussieomgeving; mail).

Reflective Practice Internship Science Communication

Course Code	AM_1163
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C.A.C.M. Pittens MSc
Examiner	dr. C.A.C.M. Pittens MSc
Teaching Staff	
Teaching method(s)	

Course Objective

The Reflective Practice Placement (RPP) internship, or the Science Communication Research Internship, is compulsory for students who wish to do a full Science Communication studypath, as part of their Master program. The aims of the RPP internship are:

- Learn to independently apply and expand your practical science communication skills in the scientific research field of your main Master program, e.g. (multi-)media content production, science journalism,

facilitation of dialogue, policy and strategy development, exhibit content design, etc.).
- Critical self-assessment and reflection on acquired science communication competencies in the field.
- Conduct scientific research into science communication practices independently: assess scientific information, design a research project, apply scientific methods, collect data, report and discuss findings.
- Present and discuss about internship and research outcomes.
- Learn to cooperate with researchers and practitioners of various disciplines.
- Gain an impression of a potential future field of career.

Course Content

When you are doing the full VU Science Communication studypath or the UvA Major Science Communication, you need to conduct one internship (30 ECTS, 5 months). One of the two possible formats is the Reflective Practice Placement (RPP) Internship.

The complete and up-to-date information about the internship can be found in the SC RPP internship guide line on Canvas (Science Communication Community page).

Additional Information Teaching Methods

Work-based placement (in an organisation that does science communication on a professional level).

Method of Assessment

Written report and oral presentation.

Within four to six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam.

The day-to-day supervision can be carried out by a trainee research assistant (AIO), postdoc, researcher or employee of a science (communication) organisation with at least a completed Master of Science. This onsite supervisor is involved in the assessment of the internship attitude and execution as well.

Entry Requirements

Only for students enrolled in a Master program of a Science Faculty.

Additional Information Target Audience

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the SC RPP Internship guidelines (on Canvas) and in the Academic and Examination Regulations of your own Master program.

Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guidelines of Earth science (on Canvas) and in the Academic and Examination Regulations.
The work-based placement is subject to the Faculty of Science document: “Student placement (internship) and literature regulations”. These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master research project. This agreement should be sent for approval by the science communication co-ordinator at least two weeks before the planned start of the work-based placement. If the proposal is of sufficient quality, you can start your internship. If not, you’ll need to adapt your proposal and send it for approval again. You can only start your internship after your research design has been approved.
The placement may be extended by 6 EC, subject to conditions that can be found in the Faculty of Science document “Student placement (internship) and literature regulations”. The student must send a request for extension to the Earth science Examination Board.
Information on Master internships is made available on Canvas.

Research Internship Science Communication

Course Code	AM_1162
Credits	30
Period	Ac. Year (September)

Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C.A.C.M. Pittens MSc
Examiner	dr. ir. M.G. van der Meij
Teaching Staff	dr. ir. M.G. van der Meij, dr. J.F.H. Kupper
Teaching method(s)	

Course Objective

The Research Internship Science Communication, or the Reflective Practice Placement (RPP) Internship, is compulsory for students who wish to do a full Science Communication studypath, as part of their Master program. The aims of the Research Internship Science Communication are:

- The student learns to independently conduct scientific research.
- The student improves their capacities regarding the independent search for scientific information and the evaluation/ assessment hereof for use in the context of their own research question.
- The student improves their ability to apply scientific methods and knowledge, to answer research questions and to generate evidence-based knowledge.
- The student improves their ability to formulate a research question, to choose, to implement and to evaluate the (appropriate) research method, and to phrase the obtained results in report.
- The student improves their ability to cooperate with researchers of various disciplines.
- The student improves their ability to orally present the research results and to discuss the findings.
- The student obtains a good impression of a potential future field of career.

Course Content

the UvA Major Science Communication, you need to conduct one internship (30 ECTS, 5 months). One of the two possible formats is the Research Internship Science Communication.

The complete and up-to-date information about the internship can be found in the SC research internship guideline on Canvas (science communication community page).

Additional Information Teaching Methods

Work-based placement (in a research organisation or institute)

Method of Assessment

Written report and oral presentation.

Within six weeks after the start of the master internship, an interim evaluation will take place to assess whether there is a reasonable chance of the placement being brought to a successful completion.

The internship is supervised and assessed by two lecturers. Both lecturers are members of the academic staff at VU University Amsterdam.

The day-to-day supervision can be carried out by a trainee research assistant (AIO), postdoc or researcher.

Entry Requirements

Only for students enrolled in a Master program of a Science Faculty.

Additional Information Target Audience

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the SC RPP Internship guidelines (on Canvas) and in the Academic and Examination Regulations of your own Master program.

Additional Information

Participation in this compulsory component is only permitted if the student meets the relevant requirements for admission. These requirements are detailed in the Internship guideline of science communication (on Canvas) and in the Academic and Examination Regulations.
The work-based placement is subject to the Faculty of Science document: “Student placement (internship) and literature regulations”. These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master research project. This agreement should be sent for approval by the science

communication internship or master co-ordinator at least two weeks before the planned start of the work-based placement. If the proposal is of sufficient quality, you can start your internship. If not, you'll need to adapt your proposal and send it for approval again. You can only start your internship after your research design has been approved. The placement may be extended by 6 EC, subject to conditions that can be found in the Faculty of Science "Student placement (internship) and literature regulations". The student must send a request for extension to the earth science Examination Board.
Information on Master internships is made available on Canvas.

Research Methods for Analyzing Complex Problems

Course Code	AM_1182
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. D.H.J. Lynch
Examiner	drs. D.H.J. Lynch
Teaching Staff	A. el Kahal, M.S. van der Loo MSc, B.R. Chakmak, E. Blokbergen, J.R.S. Tjien Fooh MSc, A.R. de Leeuw, K.E. Mujica Gastelbondo MSc, M. Fagan, L.L. Ruijsch, H.B.A. Tahboub MSc, S.S. Mokashi, D.P. Warnaars, A.E. Wilms, D.M.E. Servania MSc, J. Putter MSc
Teaching method(s)	Computer lab, Seminar, Written partial exam, Lecture

Course Objective

The objectives of this course are:

- To understand the differences between beta- and gamma research;
- To acquire insight and understanding of a real-world research process, including knowledge of the character of complex societal issues and the needs, advantages and disadvantages of real-world research;
- Being able to create an adequate research design for the investigation of a specific complex societal problem.
- To acquire insight into relevant research methods (both quantitative and qualitative) to address complex societal problems, their underlying theoretical concepts and their relative strengths and weaknesses;
- Being able to apply these various research methods in a specific societal context;
- To analyze and interpret qualitative findings;
- To acquire insight into reflexivity on the process of doing research in the real-world

Course Content

Contemporary societies increasingly face complex social problems, such as climate change, HIV/ AIDS or ethnic and religious diversity. These complex problems involve a variety of social actors: policy-makers, professionals, NGOs, industries, science and, of course, the public at large. Addressing these complex issues demands an approach that investigates, analyzes and integrates the positions and knowledge of different actors.

This course offers an (advanced) introduction to various research methods used in real world research, including questionnaires, surveys, semi-structured interviews, and focus groups. These methods are commonly used in research into complex problem contexts, communication and opportunities for intervention. Strengths and weaknesses of each research method and technique will be discussed, as well as its possibility to be applied in different societal contexts.

Additional Information Teaching Methods

Research Methods for Analyzing Complex Problems is a parttime course of eight weeks (6 ECTS). The total study time is 160 hours. Tuition methods include lectures, workgroups, workshops, group project work and self-study. The different elements have the following study time: - lectures 20 hours - workgroups and training 36 hours - examination 2 x 2 hours - project work & reading (self-study) Remaining hours Please note that attendance to the workgroup sessions is compulsory. If you miss one workgroup, with a good reason, you will receive an additional assignment. If you miss more than one workgroup session it is no longer possible to pass the project part of the course. Attendance to the lectures is highly recommended. In our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

Method of Assessment

The course grade is based on the group assignment 'research design' and two individual exams.
Individual Exams 60% of final grade
Group assignment 'research design' 40% of final grade

Literature

The literature of this course consists of selected scientific articles that are provided on Canvas, and the books:

- Verschuren, D.E. and Doorewaard, H. (2010). Designing a Research Project (2nd edition)Eleven International Publishing, the Hague. ISBN 978-90-5931-572-3.
- Gray, D.E. Doing Research in the Real World (3rd or 4th edition) Publications Ltd, United Kingdom. ISBN 978-1-4462-6019-7

An overview of additional literature per lecture will be provided on Canvas.

Additional Information Target Audience

The course ‘Research Methods for Analyzing Complex Problems’ is a compulsory course for first year master students ‘Management, Policy Analysis and Entrepreneurship in Health and Life Sciences’. This course is also a compulsory course within the Majors 'Science Communication' and 'Science in Society' of Health, Life and Natural Sciences Master programmes.

It can be followed as an optional course for other Life Sciences Master program students at the VU University, but only when taken in parallel with Analysis of Governmental Policy (AM_470571). Please contact the course coordinator prior to enrollment if you belong to this group.

Additional Information

The entire course is in English.

Please note that attendance to all workgroup sessions is compulsory.

Attendance to the lectures is highly recommended. From our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

Contact: Durwin Lynch (d.lynch@vu.nl)

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course, based on the actual number of students that start with the course. It is not possible to enrol for workgroups otherwise - unless announced via Canvas.

Research Methods for Health Sciences

Course Code	AM_1255
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. W. IJzelenberg
Examiner	prof. dr. R.W.J.G. Ostelo
Teaching Staff	dr. M.H. Donker, dr. D.R. Essink, dr. W. IJzelenberg, dr. R.M.H. Peters
Teaching method(s)	Seminar, Computer lab, Study Group, Lecture

Course Objective

Final attainment levels

After successful completion of this course the student:

1. Is able to critically reflect and discuss the need and limitations of inter- and transdisciplinary research in the field of health sciences
2. Is able to explain, examine and critically appraise the methodological issues and the results of qualitative studies, observational studies, experimental studies, systematic reviews and meta-analyses

3. Is able to select an adequate methodology, including quantitative and/or qualitative tools, to address a research question
4. Is able to develop a research proposal for an MSc Internship in Health Sciences with a sound justification (based on existing scientific evidence and/or relevant theories and concepts) for the relevance of the research question and a sound methodology that enables answering the research question thereby contributing to a solution for the health(care) problem addressed in this proposal.

Course Content

This Master course Research Methods in Health Sciences focuses on methods and techniques of scientific studies related to the broad field of Health Sciences.

The overall aim of this course is to master the skills for designing studies in the field of health sciences to a level that the students can formulate various research questions given a specific health (care) problem and design various types of public health studies that fit these research questions. Finally, students should be able to critically appraise the methodological quality of these types of studies.

This course is structured according to the research process from formulating the research question to analysing and summarising the results.

Additional Information Teaching Methods

The course comprises lectures, as well as workgroups and workshops. Students will work in groups on an assignment. The lectures will contain examples applied to the broad field of health sciences.

Credits for the course AM_1255:
Six ECTS (European Credit Transfer System) as indicated in the BETA Teaching and Examination Regulations (Master's Degree Programme- Vrije Universiteit Amsterdam).
One ECTS credit point equals 28 study hours, so this course includes 168 hours.

Additional Information Teaching Methods

- Twelve lectures (12 x 2 hours +and two reflection session (2x 2 hours)
- Seven workgroups (7 x 2 hours)
- Three workshops (3 x 2 hours)
- Assignment + self-study (120 hours)

Attendance is mandatory for workgroups and workshops. Lectures and reflection sessions are not mandatory, but we do recommend that you attend as much as possible, as the lectures are designed to help you write your assignment and the content is part of the exam.

Method of Assessment

Exam and an assignment (to be made in a team of students). Both the exam and the assignment should be sufficient i.e. at least graded 5,5.

The final grade consists of two parts: 1) a written exam (60% of your final mark), and 2) an assignment that you write together with a team of students (40% of your final mark). Both the exam and the assignment are in English. All parts must be graded 5.5 or higher. For the exam, there is one resit in the same academic year. There is one revision possibility for the assignment.

To complete this course the attendance obligations for workgroups and workshops must be met. If these conditions are not met, the final grade cannot be calculated/released.

Entry Requirements

Students should have basic knowledge of statistics and the principles and methods of epidemiological studies.

Students with a Bachelor's degree or premaster in Health Sciences and students with a Bachelor's degree “Gezondheid en Leven” from the VU, meet the entry requirements.

In general, students who meet the prerequisites as stipulated in the Education and Examination Regulations of the Faculty of Science Master's programme in Health Sciences and who have officially obtained access to the Health Sciences programme have the required prior knowledge to participate in the course.

Students from the Master’s programme in Biomedical Sciences at the VU only meet the entry requirements if they have completed the elective course Epidemiology (AB_470180).

Students with a Bachelor's degree comparable to the Bachelor's in Health Sciences VU (such as Health Sciences at another university, Human Movement Sciences etc.) should contact the course coordinator dr. Helma

IJzelenberg (h.ijzelenberg@vu.nl) before the start of the course.

Literature

Mandatory:

- ‘Doing Research in the Real World’. Gray, D.E. Publisher: SAGE Publications, 5th Edition (Published: November 2021). ISBN 987-1-5297-4243-5; ISBN 9781529742442.
- Selected literature to support lectures (links provided on Canvas). See the course’s Canvas site for a detailed description per lecture, workgroup or workshop.

Recommended:

- ‘Recommended: ‘Textbook of Epidemiology’. Bouter, L.M., Zeeger, M.P. Publisher: Wiley-Blackwell, 2nd edition; Published: January 2023. ISBN 9781119776024.

Additional Information Target Audience

Compulsory course for all students from the Master Health Sciences.

Compulsory course for all student from the Master Biomedical Sciences with specialization ‘International Public Health’, see Entry requirements.

Students with a Bachelor degree comparable to the Bachelor's in Health Sciences VU (such as Health Sciences at another university, Human Movement Sciences etc.) should contact the course coordinator dr. Helma IJzelenberg (h.ijzelenberg@vu.nl) before the start of the course.

Additional Information

The master course ’Research Methods for Health Sciences is a compulsory course for all Master students in Health Sciences as well as for Master students in Biomedical Sciences with a specialization in ‘International Public Health’, see Entry requirements.

This course will be offered for the final time within the MSc Health Sciences in the 2025–2026 academic year.

Custom Course Registration

For this course, it is *not* possible to register for workgroups or workshops via the registration module on your VU Dashboard. See Canvas for more information.

Explanation Canvas

All course information and announcements will be made available on Canvas (https://canvas.vu.nl). Students are therefore strongly advised, to check the course’s Canvas site regularly for important last-minute messages concerning the course.

Recommended background knowledge

Basic knowledge of qualitative studies is recommended.

Schoolvak Bio Ecologie en biodiversiteit

Course Code	O_ABBECBD
Credits	5
Period	Ac. Year (September)
Course Level	300
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	
Teaching method(s)	

Course Objective

Vakkennis aanvullen zodat de opleiding tot leraar Biologie kan worden voltooid.

Course Content

Deze module is bedoeld voor studenten van de VU Lerarenacademie met een deficiëntie op het gebied van de ecologie en de (Nederlandse) biodiversiteit. De student bestudeert zelfstandig een aantal hoofdstukken uit Biology – A Global Approach (12th Global Edition), en neemt vervolgens contact op met de vakdidacticus biologie voor een afspraak over het afleggen van het (online) tentamen. Daarnaast moet een portfolio samengesteld worden waarin verslag wordt gedaan van zes (veldbiologische) excursies die door de student zijn gedaan. Het portfolio wordt ter beoordeling aan de vakdidacticus gestuurd. Als de module succesvol wordt afgesloten worden 5 EC toegekend.

Additional Information Teaching Methods

Zelfstudie

Method of Assessment

Online tentamen en portfolio.

Schoolvak Bio Evolutie

Course Code	O_ABBIEVO
Credits	3
Period	Ac. Year (September)
Course Level	300
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	
Teaching method(s)	

Course Objective

Vakkennis aanvullen zodat de opleiding tot leraar Biologie kan worden voltooid.

Course Content

Deze module is bedoeld voor studenten van de VU Lerarenacademie met een deficiëntie op het gebied van de evolutiebiologie. De student bestudeert zelfstandig een aantal hoofdstukken uit Biology – A Global Approach (12th Global Edition), en neemt vervolgens contact op met de vakdidacticus biologie voor een afspraak over het afleggen van het (online) tentamen. Als de module succesvol wordt afgesloten worden 3 EC toegekend.

Additional Information Teaching Methods

Zelfstudie

Method of Assessment

Online tentamen

Schoolvak Bio Planten

Course Code	O_ABBIPL
Credits	5
Period	Ac. Year (September)
Course Level	300
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts

Teaching Staff	
Teaching method(s)	

Course Objective

Vakkennis aanvullen zodat de opleiding tot leraar Biologie kan worden voltooid.

Course Content

Deze module is bedoeld voor studenten van de VU Lerarenacademie met een deficiëntie op het gebied van de plantkunde. De student bestudeert zelfstandig een aantal hoofdstukken uit Biology – A Global Approach (12th Global Edition), en neemt vervolgens contact op met de vakdidacticus biologie voor een afspraak over het afleggen van het (online) tentamen. Daarnaast moet een portfolio samengesteld worden waarin verslag wordt gedaan van verschillende practica en onderzoekjes die door de student zijn uitgevoerd. Hett portfolio wordt ter beoordeling aan de vakdidacticus gestuurd. Als de module succesvol wordt afgesloten worden 5 EC toegekend.

Additional Information Teaching Methods

Zelfstudie.

Method of Assessment

Online tentamen en portfolio.

Science and Communication

Course Code	AM_470587
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	dr. C. Boersma MA MSc., T.F.L. Roedema MSc, L.B. Clauss MSc, K.E. De Long MSc
Teaching method(s)	Study Group, Lecture

Course Objective

- Gain theoretical insight in the nature of science,
- Gain theoretical insight in the nature of communication,
- Gain theoretical insight in the relationship between science and society,
- Gain insight in the role of science communication in this relationship,
- Acquire knowledge of different theories and models of science communication,
- Acquire knowledge of different strategies, media and activities for science communication,
- Learn how to practically apply theoretical concepts from the field of science communication in communicating science,
- Develop practical skills for science communication (especially writing and giving oral presentations).
- Reflect on your own knowledge and competencies pertinent to your projected (ideal) role as science communicator.

Course Content

Science is all around us and shapes our lives in many different ways. From the vaccines you need to get when traveling abroad to the smartphone you use on a daily basis, and from the public transportation you use to get to the university to the ingredients of your toothpaste: scientific knowledge is elemental to all of these. Simultaneously, society shapes the ways in which science and technology develop too. Science, technology and society influence each other continuously—or, to put it differently, they ‘communicate’.

Students of the Science Communication specialization are expected to become experts in understanding and designing interactions between science and society. In order to make this interaction fruitful and valuable for both science and society, it is first of all important to gain theoretical knowledge about science, about communication and about science communication. Science and Communication provides students with the theoretical and

conceptual foundations of the discipline of science communication. Thus, you will develop an in-depth understanding of communication processes at the core of several interfaces, including those between scientists from different disciplines, between different sciences and their stakeholders, and between science and the public.

Additional Information Teaching Methods

Expected contact hours:
Lectures (20 h)
Workgroups (25 h)
Home-study for group assignments (15 h)
Home-study for individual assignments/exam (100 h)

Method of Assessment

a) Participation. (pass/fail)
This consists of the following:
- the 'CARQ' presentations and summaries;
- Active participation in the work groups.
All these are assessed as pass or fail. For each one you fail, you have to do an alternative assignment.
Nota bene: if you fail your participation, this cannot be compensated with an alternative assignment!

b) An interview with a science communicator of your own choosing, including a reflection report. (20%)
d) "TED-talk" in which you present the research you did (e.g. for your Bsc thesis or (first) Msc internship) and reflect on it. (20%)
e) Exam. (60%)

To pass, your grades for the assignments and exam all need to be 5.5 or higher.

Resit: In case your grade is not sufficient, you have to take a resit. This can either consist of a second attempt at (b), (c) or (d), or a re-exam.

Literature

Academic articles. Direct links to articles will be provided on Canvas.

Additional Information Target Audience

The course Science and Communication is a compulsory course for students of the Master specialisation Science Communication (Wetenschapscommunicatie) and is a prerequisite for the internship. Science and Communication is an optional course for students from other master programs in the health and life sciences.

Science in Dialogue

Course Code	AM_1002
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E. Gommeh
Examiner	dr. E. Gommeh
Teaching Staff	mr. D.T. van der Lugt MSc, P.E. van der Aar, M.C. Vos MA MSc., J. Putter MSc, A.M. Aris, A.R. de Leeuw
Teaching method(s)	Seminar, Study Group, Written partial exam, Lecture

Course Objective

To gain knowledge of and insight into:

- the basic concepts and issues in understanding science-society interactions and interfaces, both from a science and technology studies and communication science perspective
- the nature and course of interpersonal and group communication processes relevant to the formal and informal dialogue between science and society
- the nature and form of dialogical science communication, aimed at reflective learning and mutual understanding

To acquire or improve:

- individual skills for effective interpersonal communication
- individual skills for the design and facilitation of science-society dialogue

Course Content

This course invites you to explore the democracy of science and technology with a particular focus on the communicative aspects of a fruitful science - society dialogue. Science, technology and innovation hold a great promise for the progress of our societies. Take for example the opportunities of nanomedicine for targeted drug delivery or synthetic biology for renewably energy or the recycling of plastic waste. At the same time these developments are potentially controversial. They lead to a variety of questions and concerns related to risks, benefits and wider moral issues. For example, nanotechnology may contain risks for health and environment. Synthetic biology may radically change the nature and meaning of life.

Clearly, advances in science do not always match the needs, desires and expectations of stakeholders and citizens in society. On the other hand, parts of society might not always appreciate the nature of scientific findings and what they have to offer. For a fruitful relationship between science and society – especially regarding topics that are potentially controversial – a constructive science-society dialogue is necessary.

This course offers advanced lectures on the basic concepts and theories about the dynamic relationship of science and society, the nature of societal controversies surrounding science, technology and innovation, and the basic concepts and issues of dialogical science communication: communication, learning, dialogue, understanding, controversy, and democracy. A series of seminars and group assignments present communicative tools and interaction spaces that can be used to design and facilitate science-society interactions. Through training workshops you will focus on improving your individual communication and facilitation skills.

Additional Information Teaching Methods

Lectures (14h), Workgroups (28h), Training workshops (24h), Dialogue presentations (12h), Selfstudy (remaining hours)

Method of Assessment

Group assignment (50%), Two digital exams (30%), Reflection report (20%).

All assignments must be passed (grade > 6).

Literature

Is announced on Canvas one month before start of the course

Additional Information Target Audience

Optional course in the MSc specialization Science Communication

Additional Information

Independence and a cooperative attitude is expected. Attendance to workgroup/ seminars and training sessions is mandatory.

Science Journalism

Course Code	AM_471014
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	A.M. Aris, K.E. De Long MSc, dr. C. Boersma MA MSc., T.F.L. Roedema MSc
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

After following this course, you are able to:

- Review the key changes in science journalism due to changes in the science-society relationship and digitalization of the media
- Examine contemporary issues of science journalism including the emergence of issues and publics, framing, values and emotions, expertise and evidence, uncertainty and ignorance.
- Apply an interdisciplinary toolkit to think about the interactions between science, media and the public and the role of science journalism
- Useyour skills in different science communication media, such as writing, video, and documentary to generate meaningful exchange between science and society
- Describe what the professional practice of science journalism entails

Course Content

Science journalism is needed now more than ever. Science heavily influences our society, not only through the application of its knowledge but also through its worldview. Although science and technology have shown the potential to improve our lives and living conditions significantly, they also contribute to new problems, risks and concerns. This awareness has changed the science and society relationship and put science and technology under increasing public scrutiny. In the context of the complex challenges of today, such as climate change or food security, ‘the facts’ are regularly called into question. Public trust in the solutions presented by science is under pressure. There is clear demand for the scientific world to communicate effectively with the rest of society. But how to shape the public conversation about science in society? And how do the media shape our perception and appreciation of science? What is the role of science journalism? Especially with respect to the larger transformations that are taking place in the relationship between science, media and public. Digitalization has revolutionized the media landscape. Everyone now has the means to either find or generate any information themselves. This provides new opportunities, but may also spread low-quality, sensationalized and sometimes intentionally misleading information. The challenge for science journalism today is to reinvent its practice and help shape a meaningful exchange between science and society.

In this course we review the key changes in science journalism. We examine how changes in both the science-society relationship and digitalization of the media landscape have changed the science journalism practice. We look at different roles of journalism such as reporters, storytellers, watchdogs and educators and discuss what new roles might need to be developed in light of the abovementioned changes. We analyze the communication and interaction dynamics in the current media landscape according to a set of core questions: How do issues and publics emerge? What is the role of framing? What is the role of values and emotions? What is the role of expertise, evidence? What is the role of uncertainty and ignorance? Finally, we train specific skills that you need as a science journalist, such as popular writing, making popular science videos, interviewing, and program design. Throughout the course, we work together closely with professionals from the field. Our guest speakers work as freelancer, editor or producer at diverse science media, such as newspapers, broadcasting media, online magazines and documentary film. They contribute to our course via the lectures, assignments and discussions.

Additional Information Teaching Methods

Lectures and seminars on theory and practice of science journalism and skill training (36h). Considerable time is set aside for performing science journalism in assignments (108h). All sessions indicated as training sessions in the schedule are mandatory. The assignments are assessed by lecturers and fellow students (peer-review process). Self-study (remaining hours).

Method of Assessment

Several individual assignments (60%), several small group assignments (40%).
All assignments must be passed (grade > 6).

Literature

Announced on Canvas one month before start of the course

Additional Information Target Audience

All Master students with a Beta-Bachelor degree. Students taking this course as part of their VU Science Communication Specialisation or UvA Major Science Communication will have precedence over other students. Students from other faculties and or universities need to get formal consent from the course coördinator (Frank Kupper) before enrolment.

Additional Information

The course is taught in English. Because language proficiency is crucial, there is the option to hand-in assigments in Dutch. More information: t.f.l.roedema@vu.nl.

Science Museology

Course Code	AM_470590
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.G. van der Meij
Examiner	dr. ir. M.G. van der Meij
Teaching Staff	dr. ir. M.G. van der Meij
Teaching method(s)	Seminar, Fieldwork, Study Group, Written partial exam, Lecture

Course Objective

After having completed the course, students can:

- Understand and analyse theories, concepts and working principles of science communication and (exhibit) design in the context of science museums.
- Synthesise ideas for an exhibit on the basis of theories, concepts and working principles of science communication and (exhibit) design.
- Create and conduct a qualitative research in a science museum that meaningfully furthers an exhibit design process.
- Visually and textually present analysis and (user) research-grounded exhibit design processes and outcomes.
- Reflect on working in an interdisciplinary team on an exhibit design (process and outcome) for a commissioner.

Course Content

This course is about designing an exhibit for a science museum, science centre or technology centre or museum. You will get familiar with theories of science communication in museum settings and different approaches to exhibit design & development, science communication in museum contexts, and qualitative user research.

Lecturers give insight into historical developments in exhibit design, into theories, concepts and working principles of science communication and (exhibit) design, and provide examples of transformative experiences in contemporary science centres and museums, but also touch upon the role and work of science communicators in museums, and research(ers) in the field of science museology.

Through individual and group assignments you are encouraged to combine theory and practice, working step-by-step towards an exhibit design. The design assignment is commissioned by a science museum, such as NEMO, Museon, Naturalis, Delft Science Centre, or Artis.

Additional Information Teaching Methods

- (Working) Lectures
- Workgroups
- Home-study for group assignments
- Home-study for individual assignments
- Field work (commissioner interview, museum analysis and user research)

Method of Assessment

Group assignment(s) (60%) and individual assignment(s) (40%). For all assignments a pass-grade must be obtained. There is no (traditional) exam.

Entry Requirements

Most students follow this course after having done Science and Communication and RMCP.

Yet for all two-year Master students, it is possible to follow this course as an elective (in case your TER and/or exam board allow this). In that case, additional reading may be asked from students, depending on the student's educational background. Please email the coordinator in advance.

Literature

Academic articles. Direct links to articles will be provided on Canvas before the beginning of the course.

Additional Information Target Audience

Optional course in the Science Communication master specialisation that is part of several two-year Master programs of the Faculty of Science. Master students (preferably from two-year Master programmes) from other universities in any scientific field are welcome as well. Additional reading may be required. Please email the coordinator in advance.

Additional Information

This course contains guest lecture(s) from and (an) excursion(s) to a science museum.

Recommended background knowledge

F., de Bakker, L., Dijkstra, A.M., and Jensen, E.A. (2020). Science communication, an introduction. World Scientific. Chapters 1, 2, 3, 4, 5 & 6. Hardcopy ISBN 978-981-120-987-1.

Study and Career year 1 M BMED

Course Code	AM_1258
Credits	0
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Chubar
Examiner	V. Chubar
Teaching Staff	V. Chubar, K. Wessels MSc, dr. J.I. Geertsema
Teaching method(s)	Seminar, Study Group, Other, Symposium

Course Objective

The Study & Career course intends to activate our students to thinking more professionally and in an engaged manner about their study habits, internships and future careers as biomedical scientists. This course fully runs across 2 academic years, together with the course AM_1270.

Course Content

During year 1, we start preparing you for (finding) the internship, with a dedicated first workshop and internship symposium.

All workshops and assignments in year 1 aim to provide some context, preparation & feedback on the professional skills that scientific professionals are often required to engage in when they're not in the lab.

Workshop topics include communication & feedback skills, peer-review, presentation skills, poster and scientific figure design, self-reflective practice, networking, goal setting and project planning.

We'll discuss these topics, analyze examples, and practice using both in-class activities and small assignments.

During year 2 (AM_1270) you will put what we've learned from the year 1 workshops and assignments into practice by putting together a learning portfolio with a professional development project.

As part of this course, our students are encouraged to get involved in organizing and attending our alumni career event, and a student research symposium, where student internship projects can be presented as scientific posters to peers and local academics. Students are also encouraged to think about their second year study plans.

Additional Information Teaching Methods

Year 1 active workshops, online clips and homework assigments. Workshops are given in small parallel groups.

Method of Assessment

Submitted assignments. Students are obliged to attend workshops and participate actively.

This course will be assessed with a pass/fail.

Additional Information Target Audience

Compulsory course for first year M Biomedical Sciences students

Custom Course Registration

After the course has started, students should sign up for workshop groups using Canvas, to a group that matches their own study schedule. Canvas instructions will follow at the beginning of the academic year.

Explanation Canvas

Deadlines for this course will be communicated via the course manual & using CANVAS.

Study and Career year 2 M BMED

Course Code	AM_1270
Credits	3
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Chubar
Examiner	V. Chubar
Teaching Staff	V. Chubar, K. Wessels MSc, dr. J.I. Geertsema
Teaching method(s)	Seminar, Study Group, Symposium

Course Objective

The Study & Career course intends to activate our students to thinking more professionally and in an engaged manner about their study, internships and future careers as biomedical scientists.

Course Content

The students will now put what they learned from the year 1 workshops and assignments into practice by producing their own learning portfolio.

For this portfolio, students need to justify, design, carry out, and later reflect on their own self-directed professional development project (major project). We also organize a "project inspiration session" to prepare students for designing their project. The purpose of the project is to give students the opportunity to build their CV and fill niches in their own professional development and transferrable skill building away from the lab.

Students will use input and feedback from their tutor, project mentors, peers, and other experts in the local scientific community to bring their professional development project ideas to fruition .

Topics covered during year 2 workshops include: Pitching an idea successfully, Job market orientation, Job application & CV preparation, Open Science & Data management.

As part of this course, our students are encouraged to attend the alumni career event and our student research symposium, where student internship projects can be presented as scientific posters to peers and local academics.

Additional Information Teaching Methods

Year 2 feedback workshops, self-guided project work with mentors and tutor. Workshops are given in small parallel groups.

Method of Assessment

Submitted assignments.
Students are obliged to attend workshops and participate actively.
A submitted final portfolio.
This course will be assessed with a pass/fail.

Entry Requirements

The course AM_1258 STUDY AND CAREER YEAR 1 M BMED (0EC) needs to be passed to be able to pass this course.

Additional Information Target Audience

Second year compulsory course for M Biomedical Sciences students

Explanation Canvas

Canvas instructions will follow at the beginning of the academic year.

After the course has started, students should sign up for their workshop groups on Canvas, according to their own study timetable. Deadlines for the course will be communicated via the course manual & using CANVAS.

The Human Microbiome in Health and Disease

Course Code	AM_1310
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. R. Kort
Examiner	prof. dr. ir. R. Kort
Teaching Staff	dr. D. Molenaar
Teaching method(s)	Seminar

Course Objective

After completing this four-week course (January, 2026), students will be able to:

1. Describe the structure, composition, and function of the human microbiome in different body sites (gut, vagina, mouth, skin, urogenital tract) and explain its role in human health and disease.

Demonstrated through: active participation in lectures and discussions, application in the research proposal
Timeframe/tools: during weeks 1–2, based on lecture material, provided literature and case studies

2.Interpret and critically evaluate microbiome research data and literature, including study design, methodologies (e.g. sequencing, metabolomics), and analysis approaches (e.g. correlation vs causation, AI-based methods).

Demonstrated through: incorporation of peer-reviewed references and methodological reasoning in the written proposal
Timeframe/tools: weeks 1–3, using online databases, lecture content, and feedback sessions

3. Identify potential therapeutic, diagnostic or preventive applications of microbiome science in medical, industrial or environmental domains (e.g. probiotics, dietary interventions, microbiome-based therapies).

Demonstrated through: development of an innovative and feasible microbiome-based application in the proposal
Timeframe/tools: weeks 2–4, using insights from guest lectures and scientific literature

4. Formulate a scientifically sound and original research proposal (4–5 pages) that addresses a microbiome-related problem, includes a clear research aim, an appropriate experimental or analytical approach, a proposed application, and the expected outcomes.

Demonstrated through: submission of a written proposal with abstract, problem definition, approach, application/expected outcome figures (1-3) , references (max. 15)
Timeframe/tools: proposal development in weeks 2–4, using literature, lecture input, and feedback session

5.Communicate research ideas effectively in both written and oral formats, tailored to a scientifically literate audience.

Demonstrated through: 5-minute oral pitch (5 slides), followed by Q&A; clarity of written language in proposal

Timeframe/tools: pitch prepared in the beginning of final week and presented during the final session

6.Engage in constructive peer and instructor feedback to refine scientific ideas and improve research design and communication.

Demonstrated through: active participation in mandatory plenary feedback sessions on January 16 and 23
Timeframe/tools: group sessions with peer/instructor input

7.Reflect on the societal and environmental relevance of the microbiome, including links to planetary health and climate change.

Demonstrated through: integration of these themes in discussion and, where relevant, in the proposal
Timeframe/tools: Planetarium session in the final week; lecture content; optional inclusion in proposal

Course Content

This course provides an in-depth exploration of the human microbiome and its role in health and disease. Students will acquire advanced knowledge about the communities of microorganisms that inhabit the human body, with a focus on their diversity, function, and interactions with the host. The course covers key themes such as the composition of the microbiome in different body sites (e.g. gut, vagina, mouth), the metabolic and immunological roles of microbes, and the impact of lifestyle, diet, and medical interventions on the human microbiome. Major theories and concepts discussed include host-microbe symbiosis, microbial ecology, dysbiosis, the gut-brain axis, and cause-versus-correlation in microbiome studies. Scientific research is at the core of this course. Lectures are given by researchers from academia and industry, and students will critically engage with current literature and methodologies. The course emphasizes applications of the microbiome sciences through the development of a research proposal, in which students design an innovative microbiome-based intervention or study, connecting scientific insights to challenges in health, diagnostics, food, and environmental sustainability. The course concludes with a pitch of the students' proposals and the awarding of the anual ARTIS-Micropia Microbiome Award for the most promising application of the microbiome sciences.

* EXAMPLE OF THE COURSE PROGRAM 2025 * (subject to change in 2026!!)
THE HUMAN MICROBIOME IN HEALTH AND DISEASE 2022 (JAN 7 - JAN 30, 2025)
VENUES: THE LECTURE HALL IN THE VOLHARDING AND THE PLANETARIUM OF ARTIS

WEEK 1

Tuesday, January 7 (10.00 – 12.00 AM) - intro & applications

- An introduction into the human microbiome (10.00 – 10.50 a.m.)
- Prof Remco Kort, A-LIFE, Vrije Universiteit Amsterdam and ARTIS-Micropia
- The holomicrobiome initiative (11.10 – 12.00 a.m.) - Prof Bas Teusink, A-LIFE, Vrije Universiteit Amsterdam
- Wednesday, January 8 (10.00 – 12.00 AM) – the vaginal microbiome
- The vaginal microbiome in health and disease (10.00 – 10.50 a.m.). Prof Remco Kort
- Interactions at the mucosal surface (11.10 - 12.00 a.m.). Dr. Karin StrijbisLinks to an external site., University of Utrecht
- Thursday, January 9 (10.00 – 12.00 AM) – fibers and the gut
- The gorilla gut microbiome. (10.00 – 10.50 a.m.). Prof Remco Kort
- Modulation of inflammation with gut microbiome targeted dietary intervention [recorded lecture]. (11.10 – 11.40 a.m.) Prof Justin SonnenburgLinks to an external site., Stanford University, USA.
- Gut health enhancement by eating favorable food: The GEEF study (11.40 – 12.00). Prof Remco Kort.

WEEK 2

Tuesday, January 14, (10.00 – 12.00 AM) – the gut microbiome

- The human gut microbiome 10.00 – 10.50 a.m. Dr Guus RoeselersLinks to an external site., Danone Research, Utrecht.
- Application of AI tools for the gut microbiome 11.10 -12.00 a.m.. Dr Evgeni LevinLinks to an external site., HORAIZON, Delft
- Wednesday, January 15, (10.00 – 12.00 AM) – cause or correlation?
- Correlation implies causation (but not the way you think it does) (10.00 – 10.50 a.m.). Dr Douwe Molenaar. A-life. Vrije Universiteit Amsterdam
- Inferring cause and effect from observational studies (11.10 – 12.00 a.m.). Dr Douwe Molenaar
- Thursday, January 16 (10.00 – 12.00 AM) – Research proposal session I
- Research proposal plenary feedback session I 10.00 – 12.00 AM. Prof Remco Kort

WEEK 3

Tuesday, January 21, (10.00 – 12.00 AM) – probiotics

- Probiotics: can they deliver what they promise? (10.00 – 10.50 a.m.). Prof. Bruno PotLinks to an external site., Vrije Universiteit Brussel.
- Host-microbe interactions at the mucosal surface (11.10 – 12.00 a.m.). Prof. Ger RijkersLinks to an external site., Professor, University College Roosevelt, Middelburg
- Wednesday, January 22, (10.00 – 12.00 AM) – ecology of microbial communities
- How interactions of microbial communities drive dynamics and function (10.00 – 10.50 a.m.). Dr Matti GralkaLinks to an external site.. A-LIFE. Vrije Universiteit Amsterdam.
- Eco-evolutionary interactions in the urobiome - drivers of disease and antimicrobial resistance (11.10 -12.00 a.m). Dr Marjon de Vos. Faculty of Science and Engineering. University of Groningen
- Thursday, January 23, (10.00 – 12.00 am) – Research Proposal session II
- Research proposal plenary feedback session II Prof. Remco Kort
- Sunday January 26 (24.00 pm), Deadline Research Proposal + Pitch (5 Slides)

WEEK 4

Tuesday, January 28 (10.00 – 12.00 am) – the mouth and the brain

- The oral microbiome in health and disease (10.00 – 10.50 a.m.). Prof. Egija Zaura Links to an external site. ACTA Dental Education, Amsterdam, The Netherlands
- The gut-brain-axis (11.10 – 12.00 a.m.). Prof. Sahar El AidyLinks to an external site., Faculty of Science, University of Amsterdam
- Wednesday January 29 (10.00 – 12.00 am) - Planetarium Session
- 10.00 – 10.50 Our Microbes, our Planet (Micro-organisms and Climate Change) - Prof Jef HuismanLinks to an external site., University of Amsterdam
- 11.10 – 12.00 From Milky way to Microbes – Planetarium dome performance - Milo GrootjenLinks to an external site., Head of ARTIS-Planetarium
- Thursday January 30 The Microbiome Award (sponsored by Yakult)
- Student presentations 10.00 – 12.00 [5 x (5 min presentation + 10 min Q&A)]
- Microbiome Award in ARTIS MICROPIA 12.00 p.m.
- MICROPIA VISIT (all students) 13.00 – 15.00 p.m.

Additional Information Teaching Methods

1. Lectures

Description: Interactive lectures covering the human microbiome, methodologies, applications, and societal relevance. Contact hours per week: ±4 hours (Weeks 1–3). Attendance: **Mandatory.** Details: Includes guest lectures by academic and industry experts; provides the foundation for proposal development and discussions.

2. Literature and Case Study Discussions

Description: Group-based discussions of selected scientific papers and real-world microbiome cases. Contact hours per week: ±2 hours (Weeks 1–2). Attendance: **Mandatory.** Details: Pre-reading required; focuses on interpretation and critical thinking.

3. Feedback Sessions (Plenary & Peer)

Description: Structured feedback moments to improve research proposals through peer and input of Prof Kort. Contact hours per week: ±2–3 hours (Weeks 2–4). Attendance: **Mandatory** (plenary sessions). Details: Includes formative assessment and collaborative refinement of ideas.

4. Oral Pitch Preparation and Presentation

Description: Students prepare and present a 5-minute pitch based on their proposal. Contact hours per week: ±1 hour (Final week). Attendance: **Mandatory** (for own and peers' presentations. Details: Includes Q&A and peer feedback.

5. Planetarium Session

Description: Interactive session linking microbiome science to planetary health and climate change. Contact hours: 2 hours (Final week). Attendance: **Mandatory**. Details: Stimulates reflection on the broader societal and environmental impact of microbiome research.

6. Self-Study / Independent Work

Description: Time allocated for reading, literature search, and proposal development. Contact hours: Not formally scheduled, but students are expected to invest ±10–12 hours per week. Attendance: Not applicable. Details: Essential for successful completion of written and oral course components.

Method of Assessment

1. Written Research Proposal (70%)

Description: Students submit an original and scientifically sound microbiome-related research proposal (4–5 pages), including abstract, problem statement, research aim, methodology, expected outcomes, proposed application, and references (max. 15).

Assessment criteria:

- Scientific quality and originality of the research idea
- Clarity and feasibility of the research aim and approach
- Justification of methodological choices
- Integration of literature and understanding of the microbiome
- Quality of writing and structure
- Feasibility of the application
- Individual / Group: Individual
- Assesses Learning Outcomes: 1, 2, 3, 4, 5, 7

2. Oral Pitch Presentation (20%)

Description: A 5-minute individual oral pitch (with max. 5 slides), followed by a short Q&A session. Students present their proposal to peers and instructors.

Assessment criteria:

- Clarity and persuasiveness of the presentation
- Ability to communicate complex scientific ideas effectively
- Response to questions and interaction during Q&A
- Individual / Group: Individual
- Assesses Learning Outcomes: 5, 6

3. Participation and Engagement (10%)

Description: Active participation in lectures, discussions, feedback sessions, and the planetarium session.

Assessment criteria:

- Constructive contributions to group discussions and feedback
- Demonstrated engagement with reading material and case studies
- Willingness to integrate feedback into research design
- Individual / Group: Individual
- Assesses Learning Outcomes: 1, 2, 6, 7

Entry Requirements

No specific prior course, but a basic understanding of molecular biology and microbiology is required

Literature

A collection of selected publications will be made available on the Canvas site.

Additional Information Target Audience

The target audience includes but is not limited to MSc students Biomolecular Sciences and Biomedical Sciences

Additional Information

Venue: The lecture room in 'de Volharding' in 'ARTIS' (Plantage Kerklaan 38-40, 1018 CZ Amsterdam)

Recommended background knowledge

General and Molecular Microbiology

Tweetalig Onderwijs

Course Code	O_MTTO
Credits	3
Period	P2

Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S. Donszelmann
Examiner	dr. S. Donszelmann
Teaching Staff	dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Na afloop van het vak heb je een beeld van het Tweetalig Onderwijs (TTO) binnen het Nederlandse voortgezet onderwijs en beschik je over basale vaardigheden om de Engelse taal op een zinvolle manier als instructietaal én als leertaal in te zetten.

Course Content

Op hoe langer hoe meer scholen wordt tweetalig onderwijs aangeboden (TTO). Het doel van dit type onderwijs is dat leerlingen tegelijk met vakinhoud ook Engels leren. Soms pakt dit goed uit, maar het kan ook zijn doel voorbij schieten. Hoe zit dit?

Het gebruik van de doeltaal (Engels in dit geval) leidt niet bij voorbaat tot het leren van Engels. Leerzaam inzetten van de doeltaal is gebonden aan allerlei didactische voorwaarden waarvan de docent zich vaak maar beperkt bewust is. Het gevolg kan heel goed zijn dat leerlingen bij veel vakken nauwelijks Engels bijleren ten opzichte van het Engels dat zij al beheersten, en bovendien minder vakinhoud meekrijgen dan in het Nederlands het geval zou zijn omdat de docent zich omslachtig in een vreemde taal uitdrukt. Doeltaal-Leertaal onderzoek (Dönszelmann, 2019) geeft inzicht in de vraag welk didactisch instrumentarium er nodig is om leerlingen op leerzame manier onder te dompelen in de vreemde taal, terwijl er tegelijkertijd ook vakinhoud wordt overgebracht.

Binnen het keuzevak onderzoeken we wat dit betekent voor de TTO-praktijk binnen Nederlandse VO-scholen, en hoe goed TTO-onderwijs eruit moet zien.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Kort literatuurverslag, of onderzoek binnen de eigen praktijk (als hier TTO-onderwijs voorkomt).

Entry Requirements

Didactiek 2, Praktijk 1

Literature

Wordt via Canvas bekend gemaakt.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten

Custom Course Registration

Er geldt een maximum van 5 voor bijvakstudenten van andere lerarenopleidingen.

Vernieuwende onderwijsconcepten

Course Code	O_MLVNON
Credits	3
Period	P2, P5
Course Level	500

Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. M. Meeter
Examiner	prof. dr. M. Meeter
Teaching Staff	prof. dr. M. Meeter, dr. H.B. Westbroek
Teaching method(s)	Seminar, Lecture

Course Objective

Na het volgen van dit keuzevak, kun je:

1. beschrijven hoe het onderwijs in Nederland de afgelopen eeuwen vorm gekregen heeft en hoe met het onderwijs antwoord gegeven is op de vragen: wat willen we onze kinderen leren; waartoe dient onderwijs en hoe geven we onderwijs vorm?
2. een aantal vernieuwende vormen van VO-onderwijs opsommen (traditionele vernieuwende onderwijsconcepten, zoals Montessori, Dalton, Vrije School en hedendaagse vernieuwende onderwijsconcepten zoals gepersonaliseerd leren/Kunskapsskolan, Agora) en beschrijven wat de ideeën achter de vernieuwingen zijn.
3. een eigen leervraag over vernieuwende onderwijsconcepten en/of processen die daarbij horen formuleren en vragen bedenken die je tijdens het schoolbezoek aan docenten en leerlingen wilt stellen om de leervraag te kunnen beantwoorden.
4. de eigen leervraag beantwoorden met gebruikmaking van theorie, gesprekken met docenten en leerlingen, de bijeenkomsten en het schoolbezoek en een onderbouwd antwoord geven op de vraag: Wat vind ik belangrijk bij het onderwijs dat ik verzorg? Wat spreekt mij aan?
5. een begrippenkader hanteren waarmee je kunt nadenken en praten over vernieuwingsprocessen en vernieuwende onderwijsconcepten.
6. een beredeneerd oordeel vormen en een professionele dialoog voeren over voor- en nadelen van vernieuwingen en ongeteste aannames en kritische implementatiedetails benoemen.

Course Content

In het Nederlandse onderwijs, zeker in competitieve markten als de Amsterdamse, wordt continu vernieuwd. Scholen sturen bijvoorbeeld aan op het implementeren van gepersonaliseerd leren of projectmatig werken of onderwijs in gemengde stamgroepen. Bij dit vak bezoeken we een aantal vernieuwingsscholen en analyseren we de beweegredenen en de proceskant van onderwijsvernieuwing (de implementatie).

Additional Information Teaching Methods

Werkcolleges (waarin de dialoog centraal staat en waarin je collega's uit het de scholen interviewt) en schoolbezoeken.

Method of Assessment

Analyse van een schoolvernieuwing in de vorm van een academisch essay of in de vorm van een podcast of educlip (keuze) aan de hand van wetenschappelijke- en vakliteratuur, interviews, observaties en schoolbezoeken.

Entry Requirements

Didactiek 1 en 2, Praktijk 1

Literature

Informatie over te lezen artikelen wordt gedeeld op Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Wereldburgerschap deel 1

Course Code	O_MWRLDBS_1
Credits	3

Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	D.S. Hoefakker-Laanstra, dr. I. Pauw, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

Na het volgen van dit keuzevak:

- kan je het belang van wereldburgerschap voor leerlingen, docenten en scholen onderbouwd over het voetlicht brengen.
- ken je de verschillende definities van wereldburgerschap en kan je op grond daarvan een onderbouwde eigen formulering geven.
- kan je de verschillende vormen van wereldburgerschap noemen en kan je bij de verschillende vormen leerlijnen en concrete activiteiten noemen.
- kan je aan de hand van de gekozen definitie en de methodologie van wereldburgerschap een onderbouwde leerlijn wereldburgerschap voor een school ontwerpen
- kan je in de klas in het kader van wereldburgerschap een maatschappelijke controverse les ontwerpen en uitvoeren .
- kan je in je klas met leerlingen met behulp van de dialoog maatschappelijke thema’s bespreken

Course Content

Leerlingen van nu groeien op in een gepolariseerde multiculturele omgeving in een globaliserende wereld, waarin belangrijke thema’s als klimaat, de energietransitie en ook oorlog en vrede de landsgrenzen overstijgen. De samenleving van nu vraagt nieuwe en andere competenties van leerlingen. Zo is het belangrijk, dat leerlingen in dialoog met elkaar kunnen gaan in plaats van in debat. En moeten leerlingen zich een aanpak eigen maken om de grote maatschappelijke thema’s voor zichzelf behapbaar te maken. Ook vraagt het van leerlingen om een open en onderzoekende houding ten opzichte van andere meningen en ideeën. Anders gezegd, de maatschappij vraagt om leerlingen, die zich zowel nu in hun eigen omgeving als later in de maatschappij kunnen ontwikkelen tot wereldburgers . En het is de taak van het onderwijs hen handvatten mee te geven om wereldburgers te worden. Hoe je als docent en als school dat kan doen is het onderwerp van deze specialisatie.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Verschillende toetsvormen: leerlijn Wereldburgerschap ontwerpen, controverse les ontwerpen en uitvoeren

Entry Requirements

Didactiek 1, Praktijk 1

Literature

- Wereldgericht Onderwijzen.Biesta in praktijk, B.Eidhof e.a, 2021
- Artikelen via Canvas

Wereldburgerschap Deel 2

Course Code	O_MWRLDBS_2
Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit

Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. H.R. Goudsmit, D.S. Hoefakker-Laanstra
Teaching method(s)	Seminar, Lecture

Course Objective

Doelen.

- Studenten kennen de verschillende internationale onderzoeken naar de diverse schoolsystemen, kunnen de indicatoren van de verschillende onderzoeken aangeven en zijn in staat tot het geven van een onderbouwde kijk op deze onderzoeken.
- Studenten kennen de kritische momenten in een schoolsysteem en zijn in staat deze momenten te waarderen
- Studenten zijn in staat andere onderwijssystemen dan het Nederlandse in kaart te brengen en deze te relateren aan het eigen onderwijsstelsel
- Studenten kunnen het begrip Internationalisering definieren, de mate van Internationalisering op hun eigen school in kaart brengen en een leerlijn Internationalisering voor een school ontwerpen
- Studenten kunnen hun visie op goed onderwijs verdiepen naar aanleiding van analyseren van andere schoolsystemen. Daarnaast zijn ze ook in staat een meer onderbouwde kijk op het Nederlands onderwijssysteem te geven

Course Content

Onlangs sprak een bekende Nederlandse publicist naar aanleiding van hevige discussies rond de klimaatcrisis de gevleugelde woorden: "Velen hebben een mening, slechts weinigen komt een oordeel toe." Nederlanders staan er om bekend overal een mening over te hebben en die te pas en te onpas te ventileren.

Je kan je afvragen of dat ook niet voor het onderwijs geldt. Er wordt immers veel gezegd en geschreven over ons onderwijs zonder dat het altijd duidelijk waarop het geventileerde is gebaseerd.

En hoewel wij er dagelijks deel van uit maken is het maar de vraag hoe onderbouwd onze eigen visie is op het Nederlands onderwijssysteem. Wordt onze eigen kijk ook niet teveel ingegeven door een beperkte hoeveelheid aan eigen ervaringen en missen wij niet het overzicht of de helikopter view om te komen tot een gefundeerd oordeel? Missen wij ook niet voldoende vergelijkingsmateriaal om tot een werkelijk gefundeerd oordeel te komen? Deze keuzemodule wil dan ook in de eerste plaats een stimulans zijn om tot een dergelijk verdiepte visie te komen.

Overal in de wereld tonen regeringen en politici interesse in de uitkomsten van de benchmarks, die internationaal worden georganiseerd door instellingen als de OESO (PISA; Eduaction at Glance). De uitkomsten van deze rapporten zorgen in de deelnemende landen vaak voor onderwijshervormingen. Maar het doen van internationaal vergelijkend onderzoek roept ook veel discussie op. Kan je onderwijssystemen wel vergelijken op grond van cijfers, zijn de onderzoeken wel betrouwbaar en valide, kun je onderwijssystemen uberhaupt wel met elkaar vergelijken? In dit keuzevak zullen we dan ook ingaan op de verschillende onderzoeken, onderzoeksmethodes en kritieken. Maar ook gaan we zelf (een klein) onderzoek doen naar de verschillende onderwijssystemen om te kijken of we ons een beeld kunnen vormen van andere schoolsystemen.

Dit keuzevak is enerzijds een aanvulling op deel 1 van Wereldburgerschap maar is ook als "los"keuzevak te volgen

Additional Information Teaching Methods

Werkcollege

Method of Assessment

Het schrijven van een essay

Entry Requirements

Didactiek 1 en praktijk 1

Literature

Artikelen uitgereikt op college

Recommended background knowledge

Didactiek 1 en Praktijk 1